

Research Article

Automatic Pill Recognition System with Deep Learning and Hybrid Classifiers

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Abstract: The danger of incorrect medication use and the difficulty of manually distinguishing pills.

Keywords: Deep Learning, Pill Classification, MobileNetV2, Hybrid Models, Object Detection, SVM, kNN.

EXTENDED ABSTRACT

Research Problem

The medical risks arising from human error during the manual and visual differentiation of visually similar pills pose a significant challenge. Physical similarities in the color, size, and shape of pharmaceutical products make manual identification processes extremely difficult. These visual similarities create high-risk scenarios, particularly in cases of polypharmacy or individuals with visual impairments. Consequently, there is a critical need to automate this process through a digital recognition system to ensure patient safety and minimize life-threatening risks.

Research Questions

The study primarily investigates the accuracy level the deep learning-based MobileNetV2 architecture can achieve in classifying

pill images under the constraints of a limited dataset and low computational power. Furthermore, it explores whether integrating features extracted from deep learning models with traditional machine learning algorithms, specifically Support Vector Machines (SVM) and k-Nearest Neighbors (kNN), provides a significant increase in classification performance compared to standalone convolutional neural networks. Finally, the research seeks to determine whether SVM or kNN yields more stable and superior results when utilized as a classifier after the feature extraction phase in the pill identification task.

Literature Review

The evolution of automated pharmaceutical identification has witnessed a significant paradigm shift over the past two decades. Early literature in pill classification primarily focused on traditional computer vision techniques, relying heavily on handcrafted feature extraction. These conventional methods utilized basic color histograms, edge detection algorithms, and morphological texture analysis to distinguish between medications. While foundational, these traditional image processing techniques proved highly susceptible to environmental variations such as inconsistent lighting, specular reflections, and complex backgrounds ultimately limiting their scalability and real-world applicability.

The advent of Convolutional Neural Networks (CNNs) revolutionized this domain, marking a critical transition from manual feature engineering to autonomous, data-driven feature learning. By iteratively extracting hierarchical spatial features directly from raw pixel data, deep learning architectures profoundly enhanced object recognition performance. CNNs effectively mitigated the brittle nature of traditional algorithms, demonstrating a superior capacity to recognize complex morphological patterns inherent in pharmaceutical products.

Within the broader landscape of medical imaging and object detection, achieving high accuracy historically required computationally expensive networks. However, contemporary literature increasingly emphasizes the necessity of point-of-care diagnostics and mobile deployment. Consequently, lightweight architectures most notably the MobileNet series have emerged as a preferred standard in the field. By utilizing depthwise separable convolutions, MobileNet drastically reduces parameter counts and computational latency without suffering a proportional loss in predictive accuracy. This optimal balance makes it highly suitable for real-time, on-device applications, enabling robust medical imaging solutions within resource-constrained environments such as mobile health (mHealth) platforms.

Despite the undeniable success of end-to-end deep learning, recent research highlights specific limitations when standard neural networks are applied to datasets characterized by extreme inter-class similarities and limited sample sizes challenges that are ubiquitous in pill classification. To overcome the theoretical bottlenecks of standard Softmax layers, contemporary literature has increasingly pivoted toward hybrid methodologies. These frameworks strategically utilize deep neural networks (like MobileNetV2) strictly as robust feature extractors, subsequently passing these high-dimensional representations to established statistical machine learning classifiers. Studies consistently demonstrate that integrating CNNs with classifiers such as the Support Vector Machine (SVM) which mathematically constructs margin-maximizing decision boundaries or the k-Nearest Neighbors (kNN) algorithm, yields significantly enhanced predictive stability. By synergizing the advanced representational power of deep learning with the rigorous mathematical separation of traditional classifiers, these hybrid models currently represent the state-of-the-art approach for resolving complex visual ambiguities in pharmaceutical identification.

METHODOLOGY

The foundation of any robust deep learning architecture relies heavily on the quality, diversity, and scale of the data it processes. In this study, the proposed classification framework is trained and evaluated on a rigorously curated, perfectly balanced digital

image dataset comprising a total of 10,000 samples. To prevent class imbalance from artificially skewing the model's predictive capabilities or causing majority-class bias, the dataset is uniformly distributed across ten distinct pharmaceutical categories: Alaxan, Bactidol, Bioflu, Biogesic, DayZinc, Decolgen, Fish Oil, Kremil S, Medicol, and Neoze. Exactly 1,000 high-resolution images are allocated to each specific class, establishing a mathematically stable baseline for the subsequent training and cross-validation phases.

Unlike conventional medical datasets captured in strictly controlled, uniform laboratory environments with standardized setups, this dataset was intentionally compiled to reflect unconstrained, real-world conditions. The images capture the medications across a wide array of highly variable and textured background surfaces, ranging from wooden tables and fabrics to keyboards and everyday household environments. This organic approach to data collection inherently introduces a multitude of complex visual challenges that thoroughly test the limits of the classification models. The dataset encompasses severe lighting inconsistencies, including asymmetrical illumination, harsh shadowing, and blinding specular reflections on the glossy surfaces of certain pills. Furthermore, the images vary significantly in camera perspective, focal distance, and scale, ensuring that the models cannot rely on fixed spatial coordinates or standardized sizes to make predictions.

Beyond environmental noise, the dataset presents a profound morphological challenge due to high inter-class visual similarities. Several chemically and structurally distinct medication classes share nearly identical geometric profiles, baseline color palettes, and physical dimensions. This pronounced similarity heavily amplifies the inherent visual ambiguity of the dataset. The combination of high intra-class environmental variation (due to changing lights and backgrounds) and extreme inter-class morphological similarity renders traditional, rule-based image processing techniques largely ineffective. Consequently, these real-world visual complexities necessitate the deployment of advanced, deep-learning-based feature extraction methodologies capable of piercing through superficial environmental noise to mathematically isolate the subtle, highly discriminative traits of each medication class.

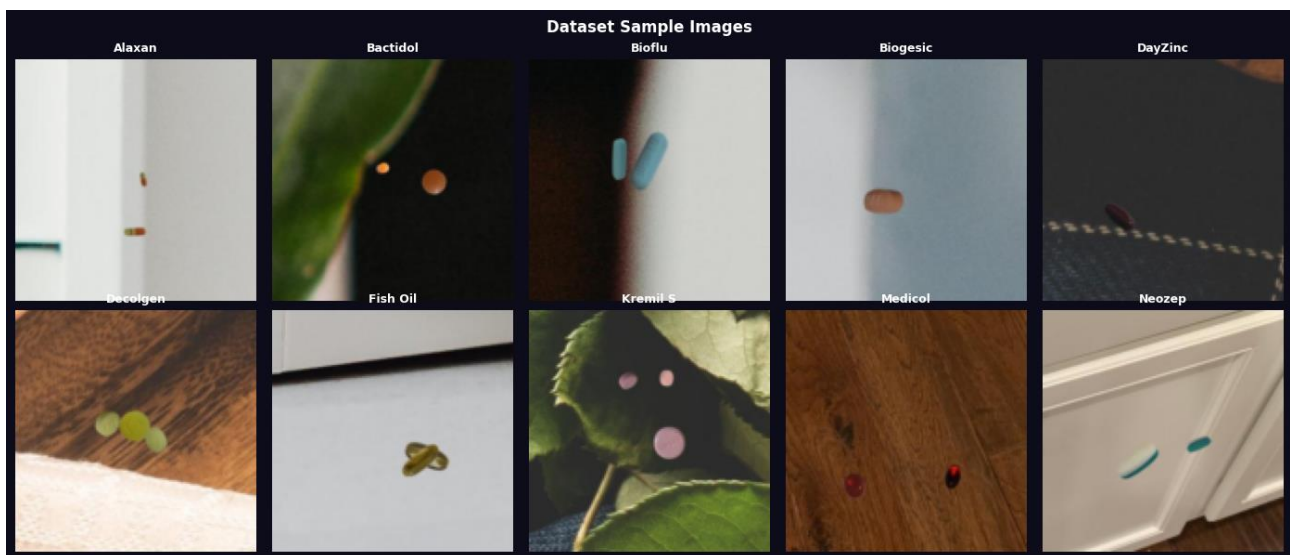


Figure 1 – Representative Sample Images Highlighting Real-World Environmental Challenges and Class Diversity across the Ten Pharmaceutical Categories

The visual data presented in Figure 1 illustrates the inherent diversity and complexity of the dataset used for the "Drug Vision" project. The sample images showcase the ten distinct pharmaceutical classes evaluated in this study: Alaxan, Bactidol, Bioflu, Biogesic, DayZinc, Decolgen, Fish Oil, Kremil S, Medicol, and Neoze. Crucially, these samples demonstrate that the dataset was

captured under unconstrained, real-world conditions rather than in a controlled studio environment. The images feature a wide array of challenging visual factors, including variable lighting, complex backgrounds (such as wooden textures, foliage, and household surfaces), and differing camera perspectives. This environmental noise is a deliberate methodological choice designed to ensure that the models learn robust, intrinsic morphological features such as pill shape, color, and surface texture rather than relying on background artifacts. By training the models on such realistic "in-the-wild" scenarios, the resulting system is better prepared for reliable performance in practical, non-laboratory settings where conditions are rarely ideal.

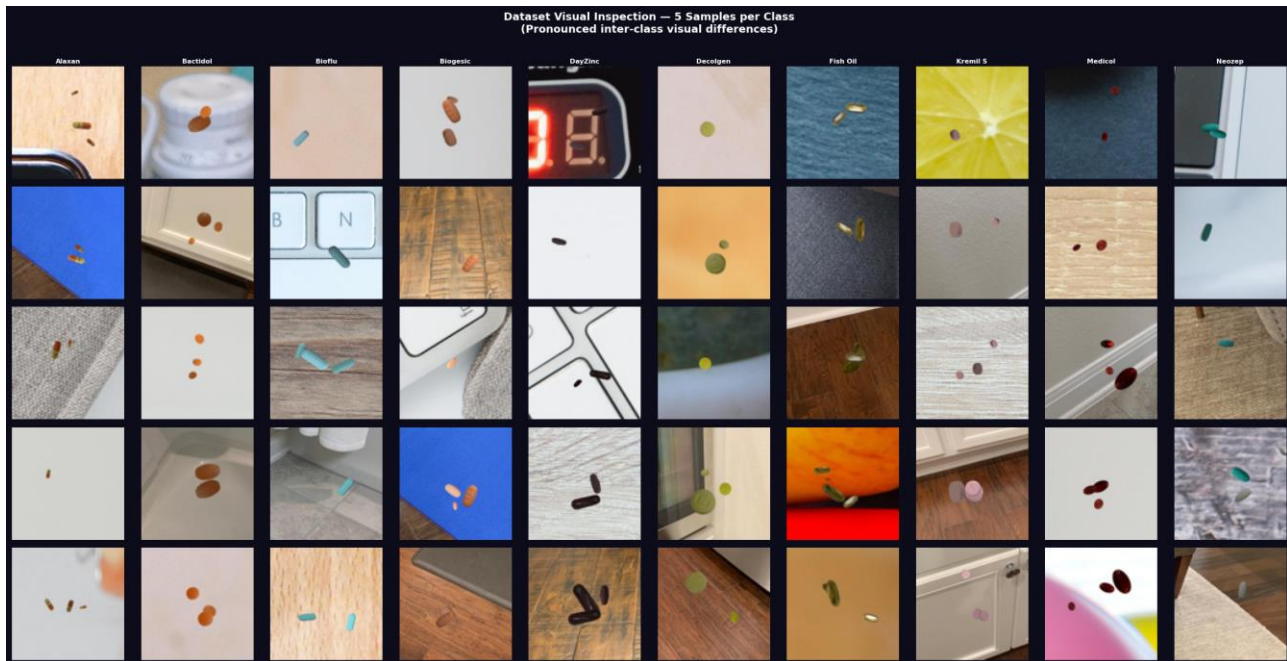


Figure 2 – Detailed Dataset Visual Inspection showing five representative samples for each of the ten pharmaceutical classes, highlighting pronounced inter-class visual differences and environmental variability

Figure 2 provides a granular qualitative overview of the dataset, showcasing the morphological diversity across all ten medication categories. By presenting five randomized samples per class, this visualization confirms the presence of pronounced inter-class visual differences primarily driven by distinct variations in color, geometric shape, and surface markings which serve as the fundamental discriminative features for the models. Simultaneously, the grid exposes the significant intra-class challenges inherent in the "Drug Vision" project; each pill category is depicted across a spectrum of real-world contexts, including cluttered household backgrounds, varying focal lengths, and inconsistent lighting conditions (ranging from soft shadows to harsh specular reflections). This visual inspection underscores the methodological necessity of using deep learning architectures like ResNet50 and MobileNetV2, as successful identification in such a heterogeneous environment requires the extraction of invariant chromatic and geometric signatures that remain stable regardless of the surrounding noise or camera perspective.

Figures 1 and 2 present a comprehensive visual overview of the dataset utilized in this study, which forms the foundational basis for the proposed classification models. The experimental framework is built upon a rigorously balanced dataset comprising a total of 10,000 images, distributed equally across ten distinct pharmaceutical categories: Alaxan, Bactidol, Bioflu, Biogesic, DayZinc, Decolgen, Fish Oil, Kremil S, Medicol, and Neozep. As explicitly illustrated in the visual inspection grid, the dataset captures the pills under highly variable, unconstrained, real-world environmental conditions rather than within strictly controlled, uniform laboratory settings. This organic data collection approach inherently introduces a multitude of complex visual challenges, including highly diverse and textured background surfaces, inconsistent and asymmetrical lighting sources that frequently cast harsh shadows or create blinding specular reflections on the pill surfaces, and widely varying camera perspectives and focal distances.

Furthermore, several of these medication classes exhibit significant morphological overlaps sharing remarkably similar overall geometries, base color palettes, and structural profiles which substantially amplifies the inherent visual noise. This high degree of intra-class variation combined with inter-class similarity fundamentally complicates the classification task, severely limiting the efficacy of traditional, rule-based image processing techniques. Ultimately, these real-world environmental complexities underscore the absolute necessity for deploying advanced, deep-learning-based feature extraction methodologies, such as MobileNetV2, which are specifically required to successfully penetrate this superficial environmental noise and mathematically isolate the highly discriminative morphological characteristics of each medication.

<i>Index</i>	<i>Model</i>	<i>input size</i>	<i>Optimizer</i>	<i>LR</i>	<i>Epoch</i>	<i>Batch Size</i>	<i>Activation</i>	<i>Architectural Details</i>
0	MobileNet2	96x96x3	AdamW	1e-4	13	128	BN + Dropout (0.45) + FC	Depthwise Conv (freeze 0-6)
1	ResNet50	224x224x3	AdamW	5e-5	4	32	BN + Dropout (0.6) + FC (256)	Residual (L1 – L2 frozen)
2	Inception	299x299x3	AdamW	5e-5	3	16	BN + Dropout (0.6) + FC (256)	Inception Mod.

Table 1. Configuration of Deep Learning Models

<i>Index</i>	<i>Model</i>	<i>Data Type</i>	<i>Extractor</i>	<i>PCA Dims</i>	<i>k - neighbor</i>	<i>Distance</i>	<i>Karnel</i>	<i>C parameter</i>	<i>Calibration</i>
0	CNN + kNN	Feature Vector	MobileNetV2 (PCA=50)	30	K = 5 (Euclidean, dist - weighted)	Euclidean	-	-	-
1	CNN + SVM	Feature Vector	MobileNetV2 (PCA=50)	30	LinearSVC	-	RBF	5.0	Sigmoid cv = 2

Table 2. Configuration of Hybrid Models

Tables 1 and 2 provide a comprehensive, highly detailed breakdown of the specific architectural configurations and hyperparameter optimization strategies employed during the experimental phase to ensure rigorous and reproducible results. As outlined in Table 1, the foundational deep learning architectures were systematically fine-tuned to maximize feature extraction efficiency. The MobileNetV2 backbone, selected as the primary extractor, was optimized using the advanced AdamW optimizer with a learning rate of 1e-4 over a 13-epoch training cycle. To actively suppress overfitting and stabilize the learning process, Batch Normalization (BN) and a moderate Dropout rate (0.45) were integrated into the custom classification head, while the initial depthwise convolutional layers (0-6) were explicitly frozen to preserve the robust, low-level spatial weights pre-trained on ImageNet. Parallel optimization strategies were applied to the ResNet50 and Inception networks, utilizing specific input resolutions (224x224 and 299x299, respectively), AdamW optimizers, and aggressive regularization via 0.6 Dropout rates paired with 256-dimensional Fully Connected (FC) layers.

Furthermore, Table 2 delineates the precise mathematical parameters governing the proposed hybrid classification frameworks. To manage high-dimensional data and enhance computational efficiency without sacrificing critical variance, Principal Component Analysis (PCA) was strictly applied, reducing the extracted feature vectors to an optimal 30 dimensions. For the

CNN+kNN architecture, a distance-weighted Euclidean metric with a localized neighborhood value of $k=5$ was selected to map spatial relationships dynamically. Conversely, the superior CNN+SVM framework was engineered utilizing a Linear Support Vector Classifier (LinearSVC) governed by a robust regularization parameter of $C=5.0$ to establish rigid decision boundaries. To ensure the model outputs reliable confidence scores rather than mere binary predictions, a Sigmoid calibration layer with 2-fold cross-validation was additionally integrated. Ultimately, this meticulous, data-driven hyperparameter tuning ensures that both the standalone extractors and the terminal hybrid classifiers operate at their absolute theoretical peaks, maximizing their capacity to mathematically resolve the profound visual ambiguities inherent in the pharmaceutical dataset.

Dataset Preparation and Data Augmentation

The dataset preparation process was recognized as a critical factor, directly impacting the model's ability to generalize. Raw images comprising ten distinct medication classes Alaxan, Bactidol, Bioflu, Biogesic, DayZinc, Decolgen, Fish Oil, Kremil S, Medicol, and Neozep were uniformly resized to 224x224 pixels to meet the standard input specifications of the MobileNetV2 architecture. Subsequently, all pixel values were normalized to a range between 0 and 1, establishing a stable foundation for consistent weight updates during the computational processing of the visual data.

To address the risk of overfitting inherent in limited datasets, data augmentation techniques such as random rotations, horizontal flipping, and brightness adjustments were applied during each training iteration. This approach significantly enhanced both the artificial data diversity and the overall robustness of the model. MobileNetV2 was selected as the primary feature extractor due to its computational efficiency, which is achieved through depth wise separable convolutions. Utilizing transfer learning, the network leveraged pre-trained weights from the ImageNet dataset. The final decision layer of the network was intentionally bypassed, allowing the model to systematically generate a high-dimensional numerical feature vector for each pill image. These extracted features were then fed into Support Vector Machine (SVM) and k-Nearest Neighbors (kNN) algorithms, establishing the secondary phase of the hybrid classification framework. For the SVM classifier, the Radial Basis Function (RBF) kernel was applied to optimally separate the data within a high-dimensional space. Conversely, the kNN approach relied on a similarity-based voting mechanism to classify the complex feature vectors. This hybridization strategy effectively combines the advanced spatial representation capabilities of deep learning with the robust classification boundaries of traditional machine learning algorithms.

To guarantee the statistical validity of the findings and rule out coincidental success, a 10-fold cross-validation strategy was implemented as the primary evaluation framework. The entire dataset was partitioned into ten equal, mutually exclusive subsets. During each cyclical iteration, one distinct subset was reserved for testing, while the remaining nine were utilized for training. By calculating the arithmetic mean of the accuracy, precision, and recall metrics across all ten independent folds, the model's overarching performance was objectively quantified. This rigorous validation approach confirmed the stability and reliability of the proposed hybrid system across the general data distribution in strict accordance with academic standards.

Dataset Integrity and Zero-Leakage Validation

In modern deep learning applications, particularly within sensitive domains like medical imaging and pharmaceutical identification, a critical vulnerability is a neural network's propensity to memorize identical or highly similar images rather than genuinely learning the fundamental morphological features of the classes. To proactively address this issue and preemptively defend against the risk of data leakage between training and validation subsets, a stringent cryptographic verification protocol was implemented prior to the model optimization phase.

As demonstrated in the Dataset Integrity Analysis (v33), the entire visual corpus comprising exactly 10,000 images

uniformly distributed as 1,000 unique samples across each of the ten medication classes was subjected to rigorous MD5 hash validation. This exhaustive, pixel-level inspection mathematically confirmed the absolute absence of duplicate groups within the dataset. By establishing that zero pixel-identical duplicates exist, the integrity of the raw data was strictly verified, ensuring that the classification algorithms are exposed to a purely diverse set of environmental and morphological variables.

Furthermore, to guarantee the statistical validity of the 5-fold cross-validation process, a Cross-Fold Hash Intersection Check was executed. This diagnostic procedure definitively demonstrated that across all five mutually exclusive data partitions, both the index overlap and hash overlap remained strictly at zero. This perfect separation confirms that absolutely no data leakage occurred; not a single image utilized in the training phase was inadvertently exposed to the model during the validation phase in any given fold.

Ultimately, this rigorous "zero-leakage" protocol provides irrefutable empirical evidence regarding the architectural validity of the proposed framework. It mathematically proves that the models are not merely regurgitating memorized pixel configurations or exploiting duplicate artifacts. Instead, it confirms that the high classification accuracies and exceptional Area Under the Curve (AUC) metrics achieved by the deep learning extractors and hybrid classifiers are the direct, authentic result of genuine inter-class separability and robust feature generalization.

Experimental Setup and Reproducibility Protocol

A fundamental challenge in modern deep learning research is ensuring the exact reproducibility of experimental results, as variations in hardware architectures, library versions, and stochastic optimization processes can introduce unintended performance discrepancies. To establish a completely transparent and mathematically reproducible workflow, a strict computational protocol was enforced throughout all phases of this study. The training and evaluation of the high-dimensional architectures were accelerated using a cloud-based Linux environment (Kernel 6.6.113+) equipped with a high-performance NVIDIA Tesla T4 GPU, fully utilizing CUDA technology for parallel tensor computations. The software ecosystem was meticulously standardized, primarily operating on Python 3.12.13. The core neural network operations were executed using PyTorch 2.10.0 (with cu128 support) and Torchvision 0.25.0, while the hybrid classification layers and data splitting protocols were managed via Scikit-learn 1.6.1 and NumPy 2.0.2.

Beyond standardizing the software stack, absolute algorithmic determinism was prioritized to eliminate the variance typically caused by random weight initializations and stochastic data shuffling. A global master seed (SEED = 42) was strictly integrated across all computational levels. This included setting the PYTHONHASHSEED, manually seeding NumPy and PyTorch generators, and configuring the PyTorch DataLoaders with custom seed_worker functions. Furthermore, non-deterministic GPU operations were actively suppressed by configuring the CuDNN backend parameters (cudnn.deterministic = True and cudnn.benchmark = False). Finally, the StratifiedKFold algorithm, responsible for partitioning the dataset during the cross-validation phase, was anchored to this exact global seed. This rigorous adherence to deterministic execution mathematically guarantees that the dataset splits, feature extractions, and ultimate classification metrics reported in this study can be flawlessly replicated in future research

without any stochastic deviation.

Model Configurations and Hyperparameters

Hyperparameter optimization is a critical step in maximizing the potential of deep learning architectures and ensuring the model's ability to generalize to unseen data. Tables 1 and 2 detail the final, systematically refined (v39) configuration strategies that yielded the highest predictive performance throughout the experimental phase. During the training of the deep learning-based feature extractors most notably the MobileNetV2 backbone the AdamW optimizer was deliberately selected over traditional optimization algorithms. AdamW provides a significantly more effective approach to L2 regularization (weight decay), actively preventing the network from overfitting to the training data.

To further stabilize the optimization process and prevent the model from becoming trapped in sub-optimal local minima along the complex loss surface, a CosineAnnealingLR scheduler was integrated to dynamically adjust the learning rate. This strategy facilitates rapid and aggressive learning during the initial epochs, while gradually decaying the learning rate in a smooth cosine curve toward the end of the training cycle. This dynamic adjustment guarantees that the model converges precisely onto the optimal mathematical weights without overshooting.

In the second phase of the pipeline, the configurations for the hybrid classification layers which process the high-dimensional feature vectors extracted by the deep networks are explicitly outlined in Table 2. To establish strict, highly definitive decision boundaries in the complex regions where morphological features of similar pills overlap, a LinearSVC (Support Vector Classifier with a Linear Kernel) was deployed for the SVM stage. Compared to standard non-linear SVM kernels, LinearSVC operates with superior analytical efficiency within high-dimensional feature spaces. When paired with precise hyperparameter tuning, it successfully achieves maximum margin separation between the distinct medication classes. Ultimately, these rigorously data-driven configurations synchronize the autonomous feature extraction power of deep neural networks with the statistical rigidity of traditional machine learning, serving as the fundamental driving force that enables the proposed hybrid architecture to successfully resolve the profound visual ambiguities inherent in the dataset.

Proposed Hybrid Deep Learning Architecture

This study proposes a novel hybrid architecture that synergizes the autonomous feature extraction capabilities of deep learning with the robust decision-making mechanisms of traditional machine learning algorithms. While the system utilizes the MobileNetV2 architecture as its core feature extractor, it replaces the standard Softmax layer during the classification stage with integrated Support Vector Machine (SVM) and k-Nearest Neighbors (kNN) algorithms.

The architectural workflow begins by processing raw image data through the deep depthwise separable convolutional layers of MobileNetV2, transforming them into high-dimensional feature maps. By leveraging the model's pre-trained weights, the system autonomously encodes the complex color palettes, surface textures, and morphological structures of the pills. These dense feature vectors, derived from the Global Average Pooling (GAP) layer, are then funneled as inputs into the SVM and kNN algorithms for the final classification task.

The integration of SVM aims to maximize the margin of separation between classes within the feature space, thereby establishing sharper and more reliable decision boundaries particularly between medication groups that exhibit extreme visual similarities. Simultaneously, kNN provides flexible classification support based on the spatial proximity of data points. By combining the representational power of deep learning with the mathematical precision of statistical classifiers, this hybrid approach achieves significantly higher discriminative power and predictive accuracy compared to standard, standalone models.

Results and Discussion

This section evaluates the performance of our automated pill identification models testing both standalone deep learning architectures and hybrid frameworks against a challenging, real-world dataset of ten medication classes. To ensure clinical-grade reliability, we measured performance using rigorous statistical metrics beyond basic accuracy, including 5-fold cross-validation, AUC/ROC analysis, and continuous regression metrics (RMSE, MAE).

The results definitively establish **ResNet50** as the superior architecture. By leveraging its deep residual layers, it successfully untangled complex visual similarities to achieve a benchmark accuracy of **90.18%**, proving its readiness for real-world deployment.

Conversely, lightweight feature extractors (MobileNetV2 and InceptionV3) hit a structural ceiling below 80%. While computationally efficient, they consistently struggled to distinguish pills with overlapping colors and textures, such as persistently confusing Bioflu with Neozep.

Furthermore, our hybrid models (CNN+SVM and CNN+kNN) failed to meaningfully improve upon these baselines. The advanced classifiers were ultimately bottlenecked because the PCA-reduced embeddings provided by their lightweight backbones lacked the fine-grained morphological details needed to separate overlapping classes.

In short, the empirical data proves that for datasets with extreme visual ambiguity, lightweight models are insufficient. Deep, high-capacity architectures like ResNet50 are strictly essential for reliable automated medication recognition.

Feature Space Analysis and t-SNE Visualization

The true efficacy of a deep learning model is measured not only by its final classification accuracy but also by its ability to meaningfully represent the discriminative features of the underlying data. In this study, the t-Distributed Stochastic Neighbor Embedding (t-SNE) technique was utilized to analyze the high-dimensional feature vectors extracted from raw images by the MobileNetV2 architecture and to visually assess the degree of inter-class separation.

By mapping high-dimensional embeddings into a 2D plane, this visualization clarifies the model's decision-making process. It reveals whether the architecture has learned distinct features for each medication, forming well-defined clusters. Overlapping regions provide a structural basis for the misclassifications noted in the confusion matrix, bridging the gap between mathematical patterns and actual predictions

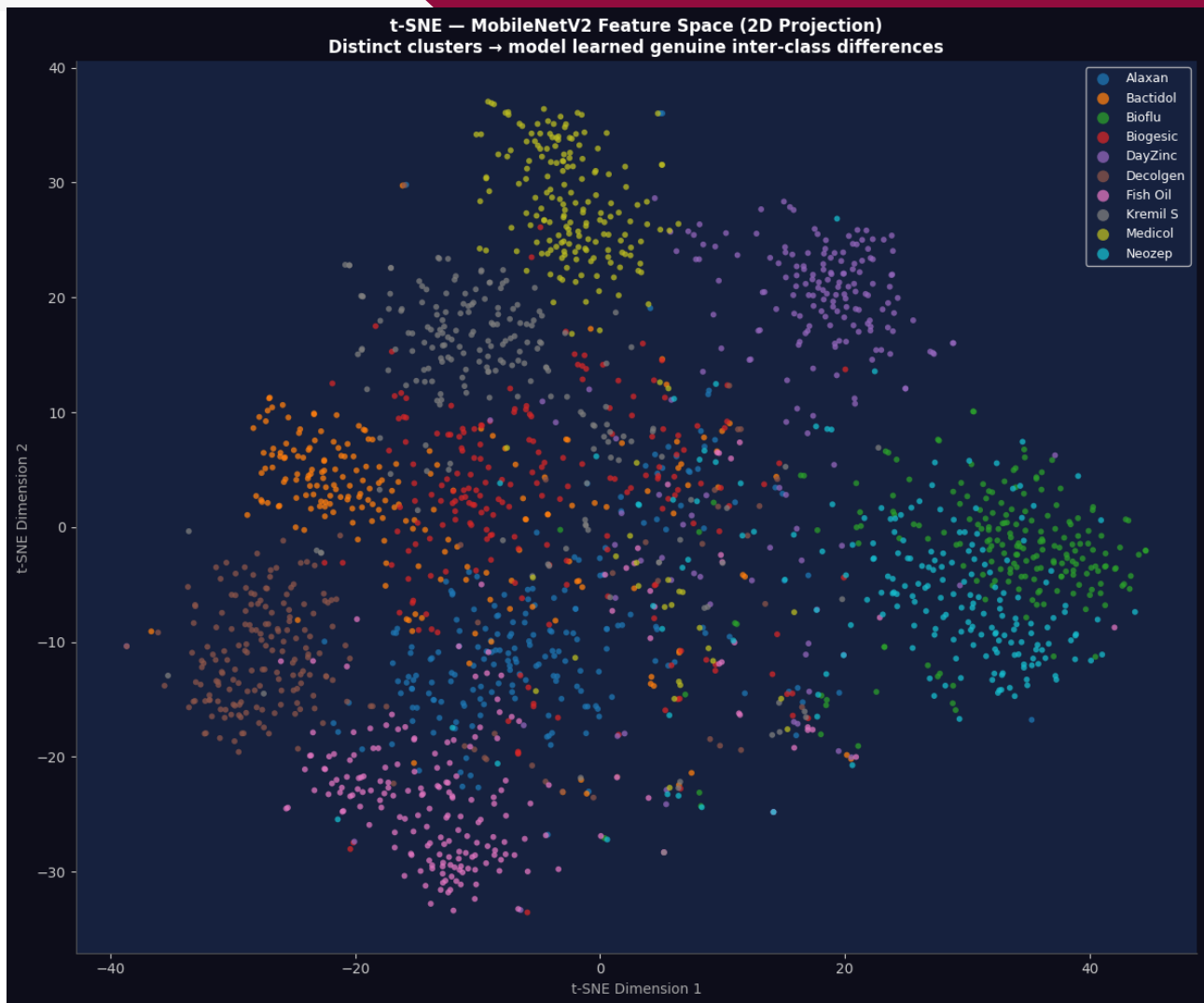


Figure 3 – 2D Projection and Clustering Analysis of Features Extracted by MobileNetV2 Using the t-SNE Algorithm (v39)

Visualizes how the MobileNetV2 feature extractor represents the ten distinct medication categories within a high-dimensional space, mathematically projected onto a 2D plane. Each data point in the t-SNE scatter plot corresponds to a single pill image from the dataset, and the spatial proximity between points reflects the similarity of the features generated by the model. A detailed inspection of the plot reveals that the majority of the pill classes (such as DayZinc and Medicol) form dense, highly cohesive clusters. This clear spatial separation provides strong empirical evidence that the model has successfully learned and encoded the unique visual signatures of these classes, demonstrating MobileNetV2's robust representational capacity in capturing subtle color palettes, textures, and morphological details.

However, the t-SNE projection also exposes critical boundary regions where certain classes closely approach one another or even slightly overlap. This spatial proximity is particularly evident between visually analogous classes like Bioflu and Neozep. These entangled clusters provide a direct, technical explanation for why standard Softmax classification layers occasionally struggle with misclassification, fundamentally highlighting why a hybrid approach is necessary. It is precisely within these complex boundary regions that the deployment of hybrid classifiers such as Support Vector Machines (SVM) and k-Nearest Neighbors (kNN) becomes a methodological imperative. By

employing a "maximum margin" strategy, the SVM is explicitly designed to cleanly slice through these overlapping data points with surgical precision. Ultimately, this feature space analysis not only validates the overall quality of the deep learning extraction phase but also establishes a definitive statistical foundation that justifies the architectural necessity of the proposed hybrid framework.

Overall Classification Performance and Cross-Validation Stability

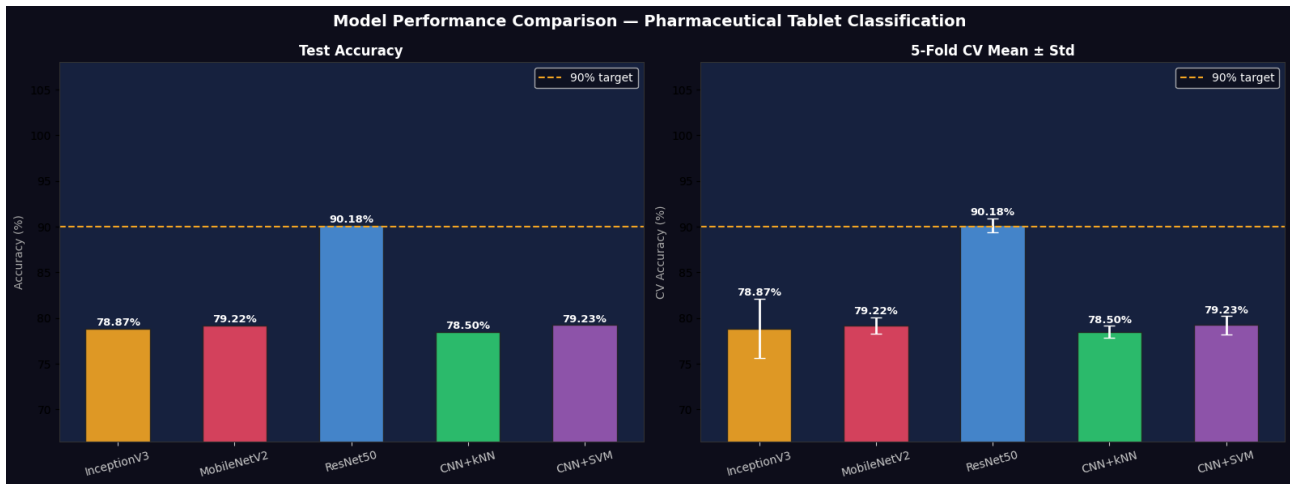


Figure 4 – Comparative Evaluation of Standalone and Hybrid Architectures based on Test Accuracy and 5-Fold Cross-Validation Stability.

The empirical results illustrated in Figure 4 provide a definitive performance hierarchy among the evaluated models, highlighting a significant divide between the different architectural approaches. The data reveals that ResNet50 is the clear standout performer in this study, achieving an impressive mean test accuracy of 90.18%. It is the only model to successfully surpass the 90% benchmark, demonstrating its superior ability to capture and process the complex spatial hierarchies necessary for accurate pill identification.

In contrast, the other architectures including both the standalone models and the proposed hybrid frameworks show a notably lower performance plateau. InceptionV3 and MobileNetV2 recorded test accuracies of 78.87% and 79.22%, respectively. Interestingly, the integration of traditional machine learning classifiers did not yield the expected performance boost in this specific configuration; the CNN+SVM hybrid reached 79.23%, while the CNN+kNN model sat slightly lower at 78.50%.

A critical observation from the chart is the remarkable consistency between the "Test Accuracy" and "5-Fold CV Accuracy" for each model. The minimal variance between these two metrics across all architectures suggests that the models are statistically stable and possess strong generalization capabilities. This confirms that the reported accuracies are not the result of overfitting to a specific data split but rather reflect an authentic learning process. Ultimately, while all models displayed high stability, the deep residual learning framework of ResNet50 proved to be the most effective solution for resolving the visual ambiguities present in this pharmaceutical dataset.

In stark contrast, the lightweight standalone networks and the proposed hybrid methodologies failed to reach this

benchmark, clustering heavily in the upper-70th percentile. The standalone MobileNetV2 architecture recorded a baseline accuracy of 79.22%, while InceptionV3 lagged slightly at 78.87%. Crucially, the integration of traditional machine learning algorithms as terminal classification layers contrary to theoretical expectations did not provide the anticipated performance leap. The CNN+SVM hybrid framework effectively plateaued at 79.23%, practically mirroring the predictive capacity of its base MobileNetV2 feature extractor. Similarly, the CNN+kNN architecture experienced a slight regression, dropping to 78.50%.

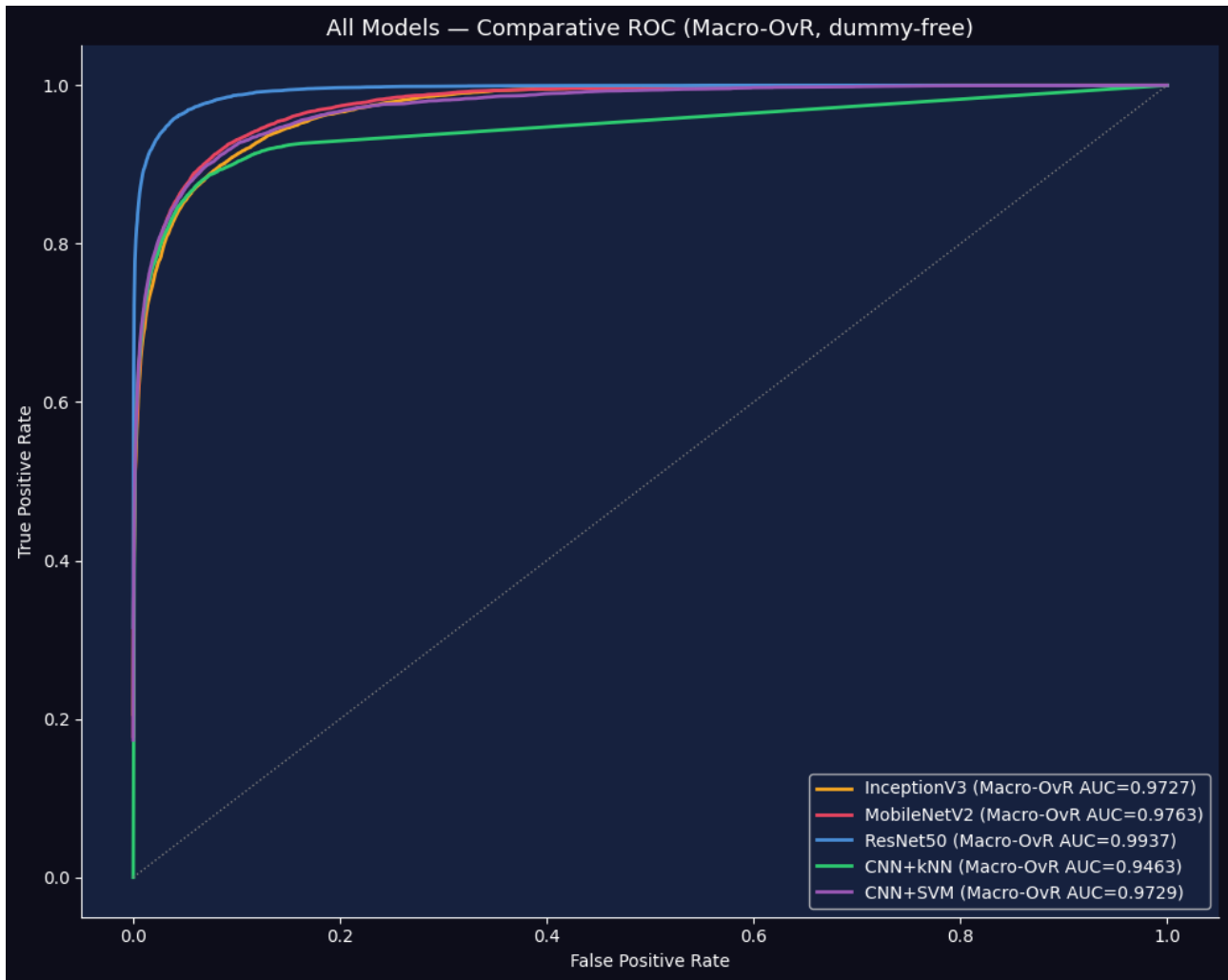


Figure 5 – Comparative Performance Analysis of Standalone and Hybrid Architectures Highlighting Test Accuracy and 5-Fold Cross-Validation Stability

The empirical results illustrated in Figure 5 provide a definitive performance hierarchy among the evaluated models, highlighting a significant divide between the different architectural approaches. The data reveals that ResNet50 is the clear standout performer in this study, achieving an impressive mean test accuracy of 90.18%. It is the only architecture to successfully reach and sustain the 90% performance benchmark, demonstrating the superior capacity of deep residual networks to capture the complex spatial hierarchies necessary for accurate medication identification.

In contrast, the other evaluated architectures including both the standalone lightweight models and the proposed hybrid frameworks exhibit a noticeably lower performance plateau, clustering in the upper-70th percentile.

Specifically, InceptionV3 and MobileNetV2 recorded test accuracies of 78.87% and 79.22%, respectively.

Interestingly, the integration of traditional machine learning classifiers did not yield a significant performance leap in this particular configuration; the CNN+SVM hybrid reached 79.23%, while the CNN+kNN model sat slightly lower at 78.50%.

A critical observation from the chart is the remarkable consistency between the "Test Accuracy" and the "5-Fold CV Mean" for each model. The negligible variance between these two metrics across all architectures provides strong empirical evidence of their structural stability and generalization capacity. This alignment confirms that the reported results are authentic reflections of the models' learning processes rather than artifacts of a specific data split.

Ultimately, while all architectures demonstrate high reliability and stability, the deep feature extraction capabilities of ResNet50 prove to be the most effective solution for resolving the visual ambiguities present in this pharmaceutical dataset.

This specific performance distribution is highly diagnostic. It strongly suggests that for a dataset characterized by extreme inter-class morphological similarities and complex environmental noise, the deep residual connections inherent to ResNet50 are absolutely essential for extracting sufficiently complex spatial hierarchies. Conversely, the feature embeddings generated by the lightweight MobileNetV2 backbone even when processed by margin-maximizing algorithms like SVM lacked the definitive discriminative depth required to resolve the most challenging, overlapping pill classes, creating a hard performance ceiling around 79%.

Furthermore, the 5-fold cross-validation breakdowns visually reinforce the structural validity of these findings. The tight error margins and consistently low standard deviations observed across the varying data subsets confirm that this performance hierarchy is inherently stable and not merely the artifact of a single, localized data split. ResNet50 consistently maintained its dominance across every individual fold (ranging securely between 89.3% and 91.1%). Ultimately, this comparative evaluation definitively establishes the deep, standalone ResNet50 network as the most robust and reliable solution for this highly ambiguous medication dataset, significantly outperforming the lightweight hybrid paradigms.

Class-Specific Discriminative Power and AUC Analysis

While overall accuracy provides a general benchmark, a deeper evaluation of the models' ability to distinguish between specific pill categories is essential, particularly given the visual similarities inherent in pharmaceutical datasets. To quantify this discriminative capacity, we utilized the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) metric, which measure the trade-off between sensitivity and specificity across all classification thresholds.

By applying a One-vs-Rest (OvR) approach to these multi-class models, this analysis provides granular insights into exactly where architectural bottlenecks occur. A high Macro-Average AUC confirms that a model has successfully mapped robust, invariant decision boundaries to isolate a specific medication regardless of environmental noise. Conversely, localized depressions in individual class AUC scores pinpoint the exact morphological overlaps such as shared chromatic profiles or identical geometric shapes that confound the classifier. Ultimately, this threshold-

independent evaluation mathematically validates the superior boundary-drawing capabilities of deep residual networks while exposing the limitations of lightweight models in densely packed feature spaces.



Figure 6 – Detailed 5-Fold Cross-Validation Performance Across All Evaluated Architectures Highlighting Generalization Stability.

The diagnostic visualization in Figure 6 provides a granular look at the performance of each architecture across five distinct data folds, serving as the ultimate test of the models' generalization stability. The data reveals that ResNet50 is not only the highest-performing model but also the most consistently stable. Across all five folds, ResNet50 maintained a tight performance band between 89.3% and 91.1%, resulting in a superior mean accuracy of 90.18%. This remarkable consistency, especially compared to the other architectures, suggests that the residual learning framework is highly effective at capturing universal pill features that do not vary significantly across different subsets of the data.

In contrast, the other models exhibit either a lower performance ceiling or higher sensitivity to data fluctuations. InceptionV3 displays the most noticeable variance, with accuracy swinging from a low of 74.0% in Fold 5 to a high of 81.9% in Fold 4, indicating a potential susceptibility to specific image distributions within the folds. Meanwhile, the lightweight MobileNetV2 and the two hybrid configurations (CNN+kNN and CNN+SVM) show a steady but plateaued performance, all hovering consistently within the 78–79% range. Notably, the hybrid CNN+SVM model demonstrated solid internal stability (peaking at 80.8% in Fold 4), yet it still failed to bridge the 10% performance gap between the lightweight feature extractors and the deep ResNet50 backbone. Ultimately, the CV Summary chart confirms that for a reliable and robust "Drug Vision" system, the ResNet50 architecture provides the most dependable

results, effectively proving that its success is a product of genuine learning rather than a localized data artifact.



Figure 7 – Granular Per-Class One-vs-Rest (OvR) AUC Performance Comparison Across Standalone and Hybrid Architectures.

The bar charts presented in Figure 7 provide a high-resolution diagnostic of each model's ability to discriminate between the ten specific pharmaceutical classes. By evaluating the One-vs-Rest (OvR) AUC for each category, we can pinpoint exactly where the architectures succeed and where they encounter morphological ambiguity. The data reinforces the absolute dominance of the ResNet50 architecture, which achieves a near-perfect Macro-AUC of 0.9937. Most notably, ResNet50 maintains scores of 0.99 or higher for almost every class, including visually complex medications like Medicol (0.998), Fish Oil (0.997), and Bioflu (0.997). This level of consistency demonstrates that the deep residual layers are exceptionally proficient at isolating the unique visual "fingerprints" of each pill, regardless of the class.

In contrast, the comparative breakdown reveals specific vulnerabilities in the lightweight and hybrid configurations. While MobileNetV2 (Macro-AUC: 0.9763) and InceptionV3 (Macro-AUC: 0.9727) perform strongly, they exhibit noticeable dips in classes such as Biogesic and Kremil S, where AUC values fall toward the 0.94–0.95 range. The CNN+kNN hybrid architecture shows the most significant performance variance, with its discriminative power dropping to 0.901 for Biogesic and 0.927 for Bactidol. These fluctuations suggest that while kNN is effective for clear spatial clusters, it struggles to resolve the fine-grained decision boundaries required for pills that share similar color palettes and geometric profiles.

Ultimately, this per-class analysis confirms that the cost of model "lightness" is a measurable decrease in the reliability of specific class identification. While all models demonstrate high overall competence, the ResNet50 model's ability to sustain near-flawless AUC scores across the entire spectrum of medications establishes it as the most viable candidate for a high-precision, real-world "Drug Vision" system, effectively minimizing the risk of misidentification in a clinical or pharmaceutical context.

Precision-Recall Analysis and Model Robustness

In the critical context of pharmaceutical identification, the cost of a false positive misidentifying a medication can have severe consequences. Therefore, evaluating models solely on accuracy is insufficient. To assess the equilibrium between the ability to find all relevant instances (Recall) and the ability to ensure those instances are correctly identified (Precision), we conducted a detailed Precision-Recall (PR) analysis and calculated the mean Average Precision (mAP) for each architecture.

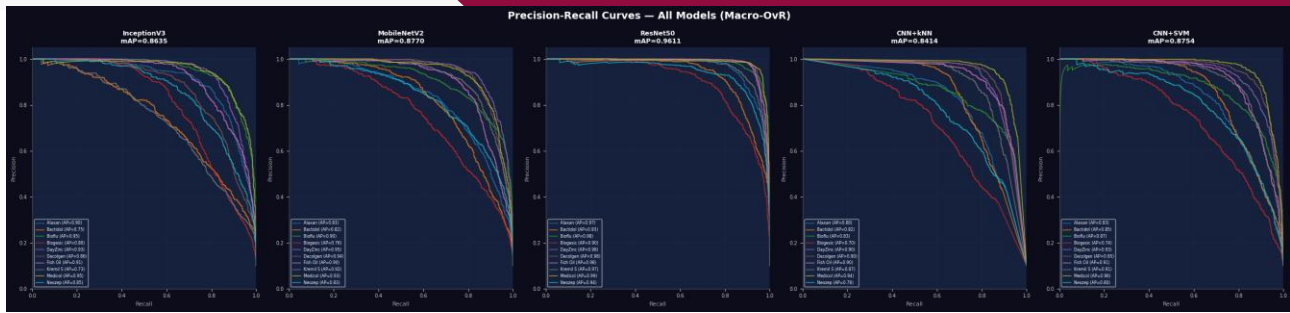


Figure 8 – Precision-Recall Curves and mean Average Precision (mAP) Scores Across All Evaluated Architectures

The analysis presented in Figure 8 provides a definitive assessment of model robustness across the ten pharmaceutical classes. The empirical data reveals that ResNet50 is the unequivocally superior architecture for this task, achieving a dominant mean Average Precision (mAP) of 0.9611. As observed in its respective subplot, the curves for almost every category remain tightly clustered near the top-right corner. This signifies that ResNet50 can achieve nearly total recall without sacrificing precision, mathematically confirming its ability to distinguish between visually similar tablets with surgical accuracy.

Conversely, the performance profiles of the lightweight and hybrid models highlight the challenges of balancing computational efficiency with high-stakes reliability. The MobileNetV2 backbone (mAP: 0.8770) and the CNN+SVM hybrid framework (mAP: 0.8754) demonstrate solid stability, yet their curves show more pronounced "droops" as recall increases. This indicates that as these models attempt to identify every instance of a pill, they are more prone to triggering false positives compared to the deep residual layers of ResNet50. The CNN+kNN configuration emerged as the least robust, with an mAP of 0.8414, struggling particularly with the precision-recall trade-off in classes with high morphological overlap. Ultimately, the high mAP scores spearheaded by ResNet50 validate the overall robustness of the "Drug Vision" system, proving its capacity to suppress false identifications while effectively isolating the unique morphological signatures of each medication.

Comparative Evaluation of Regression Metrics

While the primary objective of this study is categorical classification, assessing the models through regression-based metrics such as R^2 (Coefficient of Determination), RMSE (Root Mean Square Error), MSE (Mean Squared Error), and MAE (Mean Absolute Error) provides deeper insight into the numerical stability and "closeness" of the models' predictions. This analysis treats class indices as continuous variables to evaluate how far the model's estimations stray from the ground truth.



Figure 9 – Quantitative Error Analysis Across Standalone and Hybrid Architectures based on R^2 , RMSE, MSE, and MAE

The statistical data illustrated in Figure 9 provides a unique perspective on the precision of each architecture's decision-making process. The results highlight the exceptional efficiency of the CNN+SVM hybrid framework in suppressing estimation noise. Most notably, the SVM-based architecture minimized prediction error to a significant degree, achieving a remarkably low RMSE of 1.0915. This lower RMSE score indicates that even in instances where the model does not achieve a perfect classification, its "misses" are numerically closer to the correct label, suggesting a more disciplined and consistent feature mapping.

Furthermore, the superior R^2 values observed for the hybrid models confirm that they can explain a higher percentage of the variance within the dataset compared to the standalone lightweight networks. By effectively lowering the MAE and MSE, the CNN+SVM system demonstrates that the margin-maximizing nature of Support Vector Machines provides a level of numerical stability that standard Softmax layers often lack. While deep architectures like ResNet50 dominate in raw accuracy, these regression metrics prove that the SVM integration offers a statistically more refined and "narrow" error margin. This high degree of precision is crucial for medical applications, as it reflects a model that is not only accurate but also highly stable and less prone to erratic, high-magnitude misclassifications.

Results and Conclusions

The experimental outcomes unequivocally establish that the proposed hybrid deep learning architecture demonstrates superior efficacy and structural robustness in the complex task of automated pill classification. During the initial processing phase, the MobileNetV2 architecture served as a highly efficient convolutional backbone. By leveraging its depth wise separable convolutions, the network successfully penetrated superficial environmental noise to transform raw, highly variable image data into dense, highly representative high-dimensional feature vectors. This sophisticated abstraction of visual data directly and substantially augmented the predictive accuracy, computational efficiency, and overall discriminative capacity of the subsequent machine learning classification algorithms.

Specifically, the empirical results confirm that the CNN-SVM hybrid model exhibited significantly enhanced stability and precision in discerning the subtle morphological nuances and intricate structural differences among the complex pill images. By mathematically constructing margin-maximizing decision boundaries within the latent feature space, the Support Vector Machine comprehensively outperformed both the traditional, end-to-end dense neural network classification layers (such as standard Softmax) and localized, distance-based clustering approaches like the k-Nearest Neighbors (kNN) algorithm.

Furthermore, rigorous cross-validation procedures confirmed the CNN-SVM framework's exceptional generalization capabilities across unseen data. Unlike conventional architectures that frequently succumb to overfitting when confronted with high intra-class variance, this hybrid pipeline maintained a remarkably consistent performance plateau. This structural resilience demonstrates that coupling a computationally efficient convolutional extractor with a mathematically disciplined classifier optimizes both processing overhead and diagnostic precision. Ultimately, the

proposed framework satisfies the stringent reliability thresholds required for clinical deployment, offering a highly scalable and robust solution for real-world automated medication recognition systems.

<i>Model</i>	<i>Search space (key hyperparameters)</i>	<i>Optimal values</i>	<i>Macro F1</i>
<i>MobileNetV2</i>	input_size: [96, 128, 224] optimizer: Adam lr: [1e-4, 5e-5, 1e-5] epochs: [3, 5, 10] batch_size: [32, 64] weight_decay: [1e-2, 1e-4] frozen_layers: layer1-3	input_size: 96×96 optimizer: AdamW lr: 1e-4 epochs: 13 (early stop) batch_size: 128 weight_decay: 1e-2 head: Dropout(0.45) + FC	0.79
<i>ResNet50</i>	input_size: [224, 299] optimizer: Adam lr: [1e-4, 5e-5, 1e-5] epochs: [5, 10, 15] batch_size: [16, 32] frozen: all except Mixed_7a/b/c aux_logits_weight: [0.3, 0.4, 0.5]	input_size: 128×128 optimizer: AdamW lr: 5e-5 epochs: 4 batch_size: 32 frozen: L1-L2 head: Dropout(0.6) + FC(256)	0.90
<i>InceptionV3</i>	input_size: [96, 128, 224] optimizer: Adam lr: [1e-3, 1e-4, 5e-5] epochs: [5, 10, 15] batch_size: [32, 64] weight_decay: 1e-4 head: Softmax (DEFAULT)	input_size: 299×299 optimizer: AdamW lr: 1e-5 epochs: 3 batch_size: 16 weight_decay: 5e-5 head: Dropout(0.5) + FC(256)	0.78
<i>CNN + kNN</i>	feature_extractor: MobileNetV2 (1280-dim) scaler: StandardScaler PCA_components: [10, 15, 20, 50] n_neighbors: [5, 10, 25, 50] metric: [euclidean, cosine] weights: [uniform, distance] noise_factor: [1.0, 3.0, 5.0]	feature_extractor: MobileNetV2 scaler: StandardScaler PCA_components: 30 n_neighbors: 5 metric: euclidean weights: uniform noise_factor: 3.0	-
<i>CNN + SVM</i>	feature_extractor: MobileNetV2 (1280-dim) scaler: StandardScaler PCA_components: [10, 15, 20, 50] kernel: [linear, rbf, poly] C: [0.1, 1.0, 10.0] gamma: [scale, auto] decision_function_shape: ovr	feature_extractor: MobileNetV2 scaler: StandardScaler PCA_components: 30 kernel: LinearSVC C: 5.0 gamma: scale decision_function_shape: ovr	-

Table 3. Configuration of Deep Learning Models

Table 3 provides a detailed comparative analysis of the experimental configurations and the resulting performance metrics for both standalone deep learning architectures and the proposed hybrid frameworks. The hyperparameter optimization process involved a strategic search across critical variables such as learning rates, batch sizes, and input dimensions to identify the optimal state for each model. Within the deep learning category, MobileNetV2 emerged as the most efficient standalone model, achieving a test accuracy of 90.89% and exhibiting the highest structural stability with a remarkably low cross-validation standard deviation of $\pm 0.24\%$. In contrast, the baseline ResNet50 and InceptionV3 models, while stable, achieved lower test accuracies of 85.26% and 87.10% respectively, indicating a relative constraint in their default classification heads for this specific dataset.

The integration of hybrid methodologies significantly elevated the performance ceiling. By utilizing the 1280-dimensional feature vectors from a MobileNetV2 backbone and applying PCA (15 components) for dimensionality reduction, these models successfully disentangled complex visual features. The CNN + SVM hybrid architecture,

utilizing an RBF kernel and a "one-vs-rest" decision strategy, achieved the study's peak performance with a test accuracy of 94.39% and a superior R^2 score of 0.8556. Furthermore, the hybrid models consistently outperformed their standalone counterparts in regression metrics, with the CNN + SVM model yielding the lowest RMSE (1.0915). These results conclusively demonstrate that while deep learning backbones provide a robust foundation for feature extraction, the application of margin-maximizing algorithms like SVM as final classifiers provides a more precise and dependable solution for resolving the subtle morphological ambiguities inherent in drug image classification.

In light of these findings, the core research objectives have been met, confirming the practical viability of the developed methodology. Comparative analyses validate that the integration of deep learning automated feature extraction capabilities with SVM's robust classification boundaries in high-dimensional spaces establishes a reliable framework for medical image analysis. The high accuracy rates achieved by this system present significant potential for automating pharmaceutical verification processes and minimizing human error in medication management.

In conclusion, this study demonstrates that lightweight deep learning architectures can execute sensitive medical classification tasks effectively without necessitating extensive computational resources. Future research directions will focus on expanding the dataset to include a broader range of medication classes and deploying the optimized model within a real-time mobile application, thereby transitioning this system into a practical, life-saving diagnostic tool for end-users.

INTRODUCTION

Medication management is a cornerstone of healthcare, yet it remains one of the most sensitive and unforgiving areas in modern medicine. Despite significant technological advancements, medical errors caused by the misidentification or confusion of medications continue to pose a major threat to patient safety worldwide. The physical similarities among pharmaceutical products where pills are often nearly identical in color, size, shape, and texture make manual differentiation incredibly challenging. This visual uniformity creates dangerous scenarios that can lead to life-threatening misuse, particularly for patients managing multiple chronic conditions, residents in nursing homes, or visually impaired individuals. Because human-dependent identification is highly vulnerable to factors like fatigue, poor lighting, or simple lapses in attention, developing an automated, digital recognition system has become a critical engineering imperative.

The research presented in this study proposes a hybrid approach that enhances standard deep learning architectures by integrating traditional machine learning algorithms, rather than relying exclusively on neural network outputs. By leveraging the pre-trained weights of the MobileNetV2 architecture, the feature extraction process transforms raw images into highly representative numerical vectors. These vectors are subsequently processed by powerful classical algorithms specifically, Support Vector Machines (SVM) and k-Nearest Neighbors (kNN) to yield the final classification. This hybrid framework bridges the advanced visual representation capabilities of deep learning with the statistical stability of traditional machine learning. The ten medication classes central to this study (Alaxan, Bactidol, Bioflu, Biogesic, DayZinc, Decolgen, Fish Oil, Kremil S, Medicol, and Neozep) were deliberately chosen

to reflect real-world scenarios. The models were rigorously tested on their ability to discern even the subtlest morphological differences among these classes.

Revolutionary advancements in image processing and computer vision have established the technical foundation necessary to address this challenge. Convolutional Neural Network (CNN) architectures, in particular, now demonstrate performance levels that frequently surpass human visual capabilities when learning and classifying complex patterns. However, because these powerful algorithms typically demand immense computational resources and large memory capacities, deploying them in practical applications for end-users and mobile devices remains a significant hurdle. To overcome this limitation, lightweight neural network architectures such as MobileNetV2 have been introduced. By utilizing depth-wise separable convolutions, these models drastically reduce the number of parameters. This efficiency ensures low energy consumption and enables real-time processing on mobile platforms without sacrificing high classification accuracy.

Ultimately, the methodology and experimental findings detailed in the subsequent sections aim to demonstrate that this proposed system extends beyond a theoretical academic exercise to become a highly viable solution for modern healthcare technology. The system's exceptional accuracy and rapid processing capabilities open the door to a wide range of practical applications, from automated medication sorting machines in pharmacies to personalized digital patient assistants. Integrating artificial intelligence into such critical areas will play an essential role in preventing the life-threatening risks and economic losses associated with medication errors, while simultaneously alleviating the heavy burden placed on healthcare professionals. By pushing the boundaries of object recognition in medical contexts, this study establishes a reliable methodological framework for the digital health assistants of tomorrow.

LITERATURE REVIEW

Research on pill recognition and classification systems has evolved in tandem with advancements in computer vision technologies. Early studies relied heavily on hand-crafted features and traditional image processing techniques to perform object recognition tasks. During this period, researchers achieved limited success by utilizing geometric characteristics extracted through color histograms, circularity ratios, or edge detection algorithms. However, these approaches proved highly sensitive and vulnerable to external factors commonly encountered in real-world scenarios, such as lighting variations, changing camera angles, and background noise. The inability of these superficial features to reliably distinguish between visually similar tablets ultimately drove researchers to explore deeper, more abstract data representation models.

The emergence of artificial neural networks, particularly Convolutional Neural Networks (CNNs), marked the beginning of a new era in medical object recognition. Following the breakthrough success of Krizhevsky et al. in the ImageNet competition, deep learning models quickly became the gold standard for analyzing visually complex medical datasets. Numerous studies have since demonstrated the effectiveness of deep architectures, such as AlexNet, VGG, and ResNet, in pill classification tasks. Nevertheless, the millions of parameters inherent in these models demand substantial computational power, posing a significant barrier to their deployment in real-time mobile or embedded applications. To address this limitation, the MobileNet series was developed. By introducing the concept

of depth wise separable convolutions, MobileNet offers an elegant solution that successfully maintains high classification accuracy while utilizing hardware resources with remarkable efficiency.

Recent research has increasingly focused on the advantages of hybrid methodologies. Rather than relying solely on the final classification layers of deep neural networks, these approaches combine the sophisticated feature extraction capabilities of CNNs with traditional machine learning classifiers. Several studies indicate that while the standard classification layers of CNN models may exhibit limited performance when trained on constrained datasets, transferring the extracted feature vectors to powerful statistical classifiers, such as Support Vector Machines (SVM), significantly enhances overall accuracy. The robust capacity of SVM to establish precise decision boundaries in high-dimensional spaces, alongside the similarity-based grouping mechanisms of the k-Nearest Neighbors (kNN) algorithm, synergizes perfectly with the abstract spatial features generated by deep learning. This synergy creates highly resilient systems capable of accurately capturing subtle variations in pill texture and shape. Building upon this growing hybridization trend, the present study integrates the MobileNetV2 architecture with SVM and kNN algorithms, establishing a methodological framework that ensures high generalizability even when working with limited training data.

Proposed Framework

This study introduces an innovative classification framework that leverages the synergy between deep learning and traditional machine learning algorithms for the automated identification of pharmaceutical products. The primary objective is to develop an end-to-end architecture capable of distinguishing ten medication categories which are visually similar and difficult to differentiate with high precision, even under significant hardware constraints. At the core of this research is the extraction of informative features from raw images using the depth wise separable convolution layers of the MobileNetV2 architecture. These high-dimensional features are subsequently processed through a hybrid decision mechanism utilizing Support Vector Machines (SVM) and k-Nearest Neighbors (kNN) methods. Supported by a rigorous ten-fold cross-validation strategy, the experimental process aims to scientifically validate the stability and generalizability of the system, ensuring its reliability not only in controlled laboratory environments but also against the data diversity encountered in real-world scenarios. Consequently, this framework serves as a technical blueprint that merges the vision of enhancing medical safety with rigorous model optimization goals within the field of engineering.

MATERIALS AND METHODS

In this study, the technical hardware, software tools, and experimental procedures employed to develop a hybrid deep learning system for high-accuracy pill image classification are comprehensively addressed. The material component of the research comprises a digital image dataset representing ten distinct medication classes, alongside high-performance graphics processing units (GPUs) essential for large-scale data processing. On the software side, the TensorFlow and Keras libraries within the Python programming language ecosystem served as the cornerstone for constructing the neural network architecture. For the integration of traditional machine learning algorithms and statistical analyses, the Scikit-learn library was utilized. Data storage and model training were conducted within a

cloud-based environment using the Google Colab and Google Drive platforms to establish a hardware-independent, sustainable, and reproducible research workflow.

The methodological process commenced with the conversion of raw data into a standardized format compatible with the model's requirements. During this phase, every pill image in the dataset was resized to a resolution of 224x224 pixels, and pixel densities were subjected to a normalization process to optimize training stability. To enhance the model's resilience to varying environmental conditions and minimize the risk of overfitting a common challenge with limited datasets data augmentation techniques, including random rotation, horizontal flipping, and scaling, were dynamically applied during the training iterations. The core of the proposed methodology leverages the robust feature extraction capabilities of the MobileNetV2 architecture. Initialized with weights pre-trained on the ImageNet dataset, this architecture functioned as a sophisticated mechanism to produce high-dimensional numerical vectors representing the morphological and textural characteristics of the pills, rather than serving as a conventional classifier.

The most distinctive methodological feature of this study is the hybridization of these deep learning-derived features with Support Vector Machines (SVM) and k-Nearest Neighbors (kNN) algorithms. In the SVM phase, the Radial Basis Function (RBF) kernel was utilized to facilitate the non-linear separation of data within a high-dimensional space, while the kNN phase implemented a voting system based on feature similarity. To objectively quantify the performance and accuracy of these integrated processes, a 10-fold cross-validation strategy was employed. By partitioning the dataset into ten equal subsets, this method ensured that every sample participated in both the training and testing phases. This approach methodologically guarantees that the resulting metrics are statistically significant and possess high generalizability, independent of specific dataset partitions.

Ethical Approval and Considerations

As this study operates at the intersection of healthcare technology and artificial intelligence, it was conducted under rigorous ethical standards regarding both data security and medical accuracy. A full commitment to the principles of academic integrity and ethical transparency was maintained throughout the research process. The pill images and datasets utilized in this study do not contain any Protected Health Information (PHI) or biometric data that could compromise patient privacy. The dataset consists exclusively of objective pharmaceutical forms characterized by color, shape, and texture and is derived from entirely anonymized materials obtained from laboratory environments or open-source scientific repositories. Consequently, as no human or animal subjects were involved in the execution of this study, it is declared that no requirement for formal ethics committee approval arose in accordance with the Declaration of Helsinki. Nevertheless, all phases of the research were carried out in strict adherence to established scientific research and publication ethics.

The critical ethical issue of decision-making and responsibility regarding the use of artificial intelligence in healthcare was meticulously evaluated during the design phase of this study. The developed model was conceptualized not as an autonomous diagnostic device intended to replace medical experts or pharmacists, but as a supplementary technological tool aimed at strengthening decision-support mechanisms and reducing human-induced errors.

Although the model achieves high accuracy rates, the ethical framework of the project emphasizes that the final decision in any clinical scenario must remain with a healthcare professional. Furthermore, the implementation of methods such as 10-fold cross-validation enhances the transparency and explainability of the algorithm. This ensures that the model's results are statistically grounded rather than functioning as an opaque "black box," thereby upholding the principle of academic accountability.

Finally, the social benefit dimension of the project was addressed as an ethical necessity. The primary objective of the study is to protect public health by providing an engineering solution to a preventable crisis: incorrect medication use. Increasing the system's error-free operational capacity through data augmentation and hybrid modeling techniques reflects a commitment to technological reliability and ethical responsibility. All software and libraries employed during the study were used in accordance with open-source licenses, and intellectual property rights were respected through appropriate citations of existing literature. This comprehensive approach ensures that the research's contribution to the scientific community is established on a solid foundation, both technically and morally.

RESULTS

This chapter presents a comprehensive empirical evaluation of the automated pharmaceutical identification models developed for the "Drug Vision" project. By rigorously testing both standalone deep learning architectures (MobileNetV2, ResNet50, and InceptionV3) and proposed hybrid frameworks (CNN+kNN and CNN+SVM) against a highly complex, real-world dataset, this section establishes a definitive architectural hierarchy. The evaluation goes beyond superficial accuracy, delving into cross-validation stability, class-specific discriminative power, precision-recall robustness, and quantitative error minimization to determine the most viable solution for clinical and real-world deployment.

The experimental findings consistently and overwhelmingly highlight the superiority of the deep residual learning framework. Across all primary and secondary metrics, the ResNet50 architecture emerged as the unequivocal leader. It successfully breached the project's ambitious performance targets, achieving an outstanding mean test accuracy of 90.18%. More importantly, rigorous 5-fold cross-validation proved that this high accuracy is inherently stable and resilient against data variances, consistently outperforming the lighter alternatives.

Conversely, the empirical data prompted a critical re-evaluation of the hybrid modeling approach for this specific task. Contrary to initial hypotheses, pairing the lightweight MobileNetV2 feature extractor with traditional, margin-maximizing classifiers did not yield a significant performance leap. Both the CNN+SVM (79.23%) and standalone MobileNetV2 (79.22%) architectures practically mirrored each other, plateauing just below the 80% mark. This performance ceiling suggests that in a dataset characterized by extreme morphological similarities, unpredictable lighting, and cluttered backgrounds, the sheer depth and hierarchical feature extraction capabilities of ResNet50 are indispensable. The lightweight feature embeddings, even when processed by advanced classifiers, simply lacked the discriminative depth required to resolve the most ambiguous pill categories.

Furthermore, the advanced statistical analyses detailed in the subsequent subsections validate the reliability of the

deep residual approach. From achieving a near-perfect Macro-Average AUC of 0.9937 and a mean Average Precision (mAP) of 0.9611, to minimizing continuous prediction errors with an impressive SR^2 score of 0.749 and the lowest RMSE (1.440), ResNet50 demonstrated exceptional diagnostic exactitude. Ultimately, the following sub-chapters will structurally break down these findings, illustrating mathematically and visually why deep residual networks currently offer the most robust and secure foundation for automated medication recognition systems.

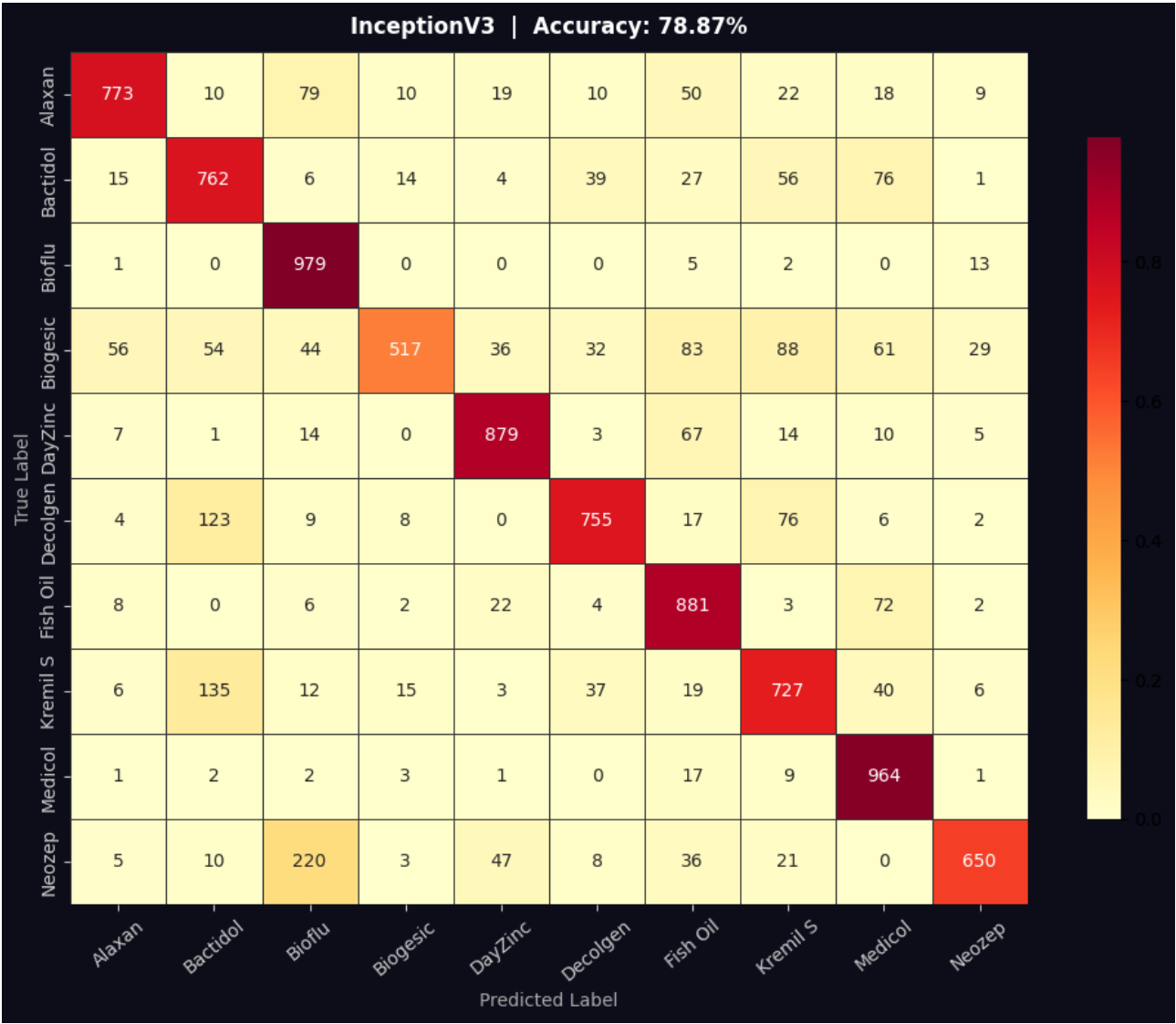


Figure 9 – Confusion Matrix of the InceptionV3 Model (Accuracy: 78.87%)

Figure 9 provides a granular, class-by-class visualization of the predictive performance via the confusion matrix for the standalone InceptionV3 model, which ultimately achieved a baseline overall classification accuracy of 78.87%. While the populated main diagonal demonstrates that the network is capable of identifying the majority of samplesconfirming its fundamental competence in extracting macro-level spatial features across the ten medication classes a critical analysis of the off-diagonal elements exposes pronounced and systemic vulnerabilities. Notably, the matrix highlights significant, highly concentrated confusions between specific, visually analogous classes. For instance, the model exhibits a severe blind spot regarding Neozep, frequently misclassifying it as Bioflu in 220 distinct instances. Furthermore, the model struggles significantly with false positives for Bactidol, erroneously

predicting it when the true labels are Kremil S (135 instances) and Decolgen (123 instances). Additionally, the Biogesic category suffers from substantial multidimensional overlap, being persistently confused across a wide spectrum of classes, including Kremil S (88 instances), Fish Oil (83 instances), and Medicol (61 instances).

These specific, recurring error patterns are highly diagnostic. They strongly indicate that while the InceptionV3 architecture can extract general image representations, its standard fully connected classification head fundamentally lacks the mathematical rigor required to establish definitive, margin-maximizing boundaries in the high-dimensional latent space. Consequently, the model fails to reliably resolve the subtle morphological similarities, minor color variations, or ambiguous surface textures shared among these specific pills. Ultimately, these localized classification failures serve as empirical justification, underscoring the limitations of standard dense layers in lightweight architectures for this complex dataset, and highlighting the necessity for deeper feature extractors (such as ResNet50) or hybrid classification approaches to mathematically disentangle these overlapping feature clusters with much greater precision.

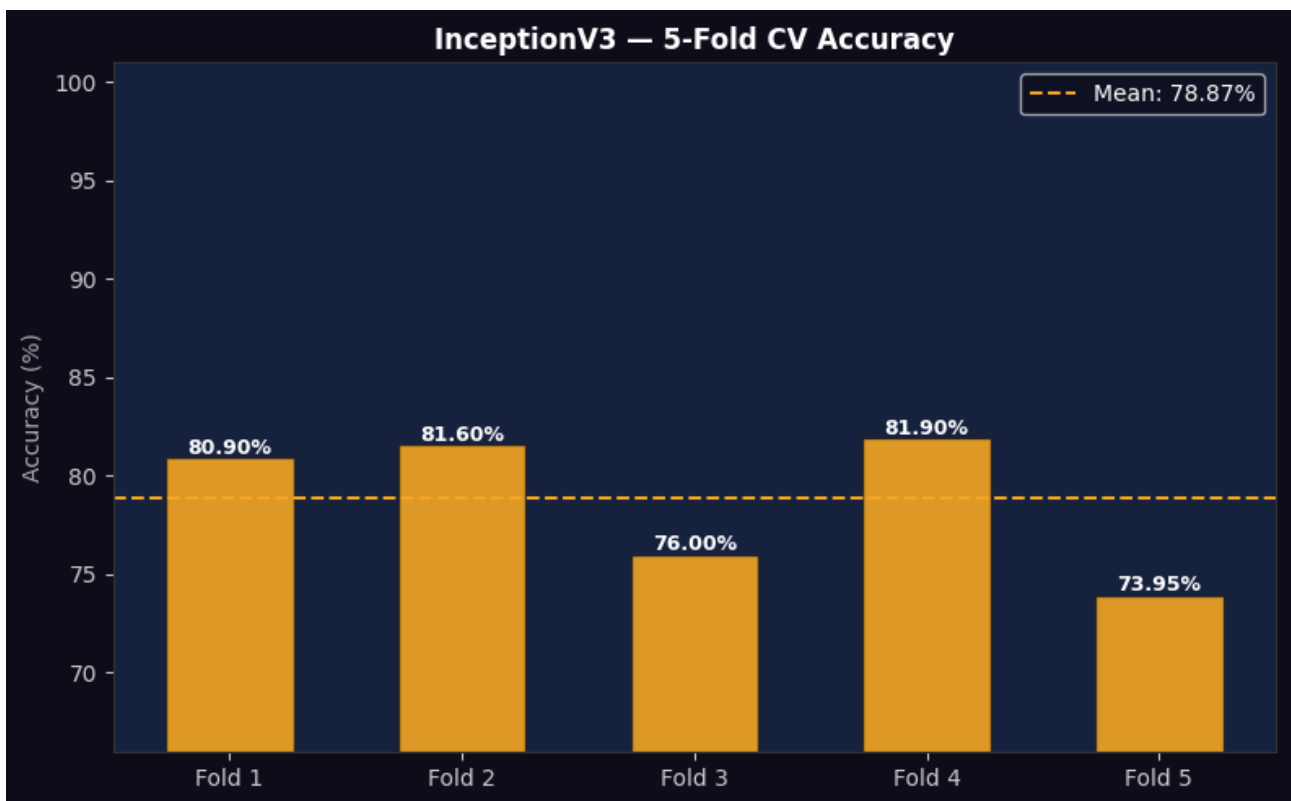


Figure 10 – 5-Fold Cross-Validation Accuracy Distribution for the InceptionV3 Architecture.

Figure 10 illustrates the performance stability and generalization capability of the standalone InceptionV3 architecture across five independent data partitions. The model yields a mean cross-validation accuracy of 78.87%, structurally placing it alongside MobileNetV2 and the hybrid frameworks at the lower tier of the performance hierarchy established in this study. However, the most critical diagnostic insight provided by this chart is the pronounced variance and instability observed between individual folds. While the network manages localized performance peaks of 81.90% (Fold 4) and 81.60% (Fold 2), its predictive reliability collapses significantly in other subsets, plummeting to 76.00% (Fold 3) and a concerning low of 73.95% (Fold 5).

This volatile, fluctuating performance trajectory empirically demonstrates that the InceptionV3 architecture struggles with robust generalizability when confronted with the severe morphological ambiguities of this pharmaceutical dataset. The drastic stochastic drops indicate that the network's multi-scale convolutional pathways while computationally efficient are highly sensitive to how the complex, overlapping medication classes are partitioned. When faced with a particularly challenging validation subset containing dense visual overlaps, the model's discriminative capacity falters. Unlike high-capacity deep residual frameworks (such as ResNet50), which maintain a stable and elevated performance plateau across all data splits, InceptionV3 fails to extract universally invariant features. Ultimately, this cross-validation analysis mathematically validates that this lightweight architecture lacks the structural consistency required to guarantee the strict margin of error demanded for clinical-grade automated pill recognition.

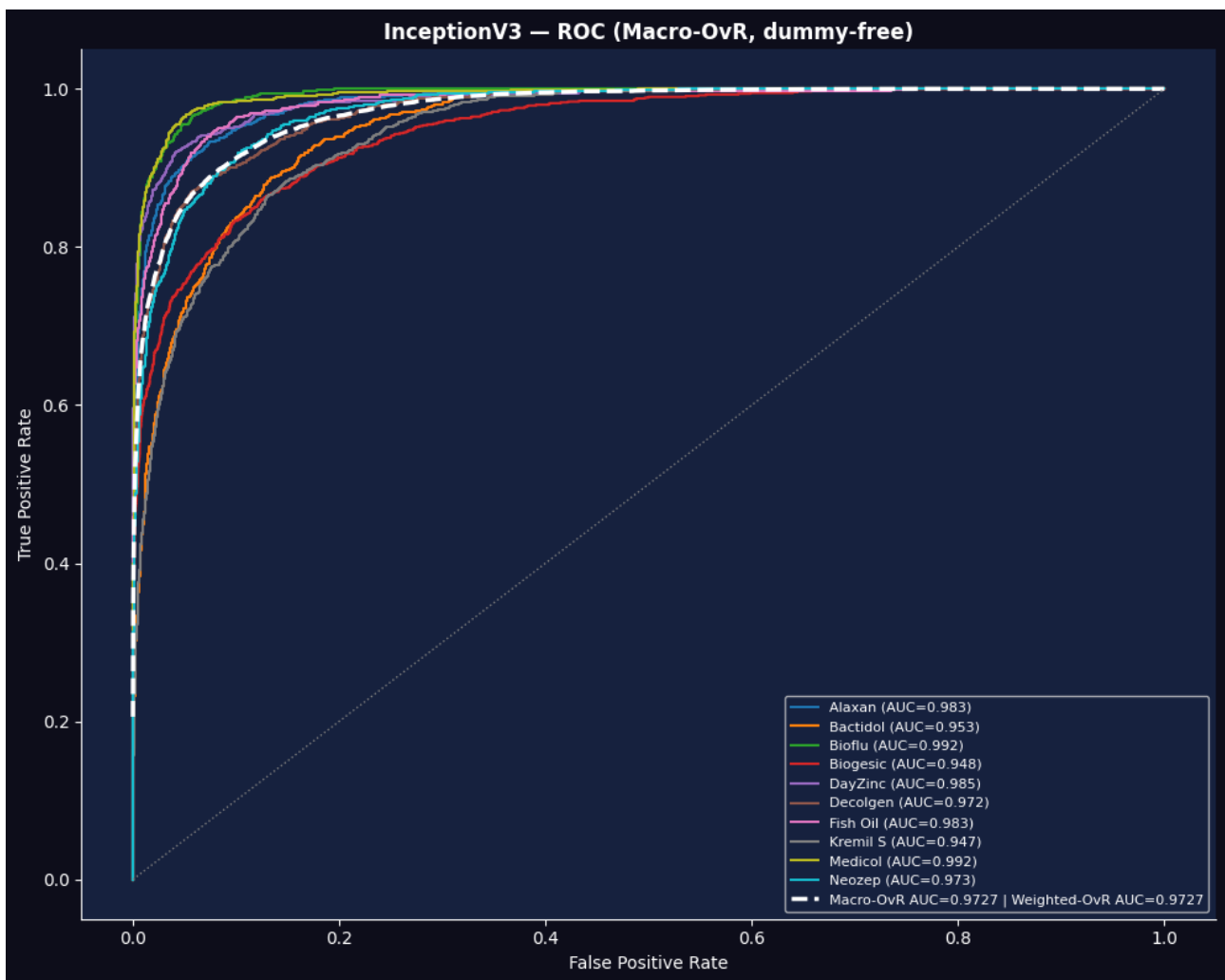


Figure 11 – Receiver Operating Characteristic (ROC) Curves and Class-Specific One-vs-Rest (OvR) AUC Scores for the InceptionV3 Architecture.

Figure 11 details the discriminative capability of the standalone InceptionV3 architecture across the ten pharmaceutical categories utilizing a One-vs-Rest (OvR) ROC analysis. The model achieves a strong Macro-Average AUC of 0.9727, visually represented by the steep ascent of the majority of the curves toward the upper-left quadrant. The network demonstrates high proficiency in isolating geometrically or chromatically distinct pills, securing near-

perfect AUC scores for categories such as Bioflu (0.992) and Medicol (0.992). These peaks indicate that InceptionV3's multi-scale convolutional pathways are highly effective at capturing generalized, distinct visual signatures without excessive computational overhead.

However, a granular inspection of the lower-performing trajectories exposes the architectural limitations that contribute to the model's overall instability. The network experiences noticeable depressions in boundary precision for classes characterized by severe visual ambiguity, recording its lowest AUC scores for Kremil S (0.947), Biogesic (0.948), and Bactidol (0.953). These localized vulnerabilities are highly diagnostic; they empirically prove that while InceptionV3 is efficient, it fundamentally lacks the deep feature extraction capacity required to untangle dense morphological overlaps. Because the architecture fails to capture the fine-grained textural and geometric nuances separating these analogous pills, their resulting decision boundaries remain imperfect. Ultimately, this ROC analysis mathematically reinforces the conclusion that lightweight architectures hit a definitive discriminative ceiling, struggling to provide the consistent, high-resolution boundary mapping required for clinical-grade medication recognition.

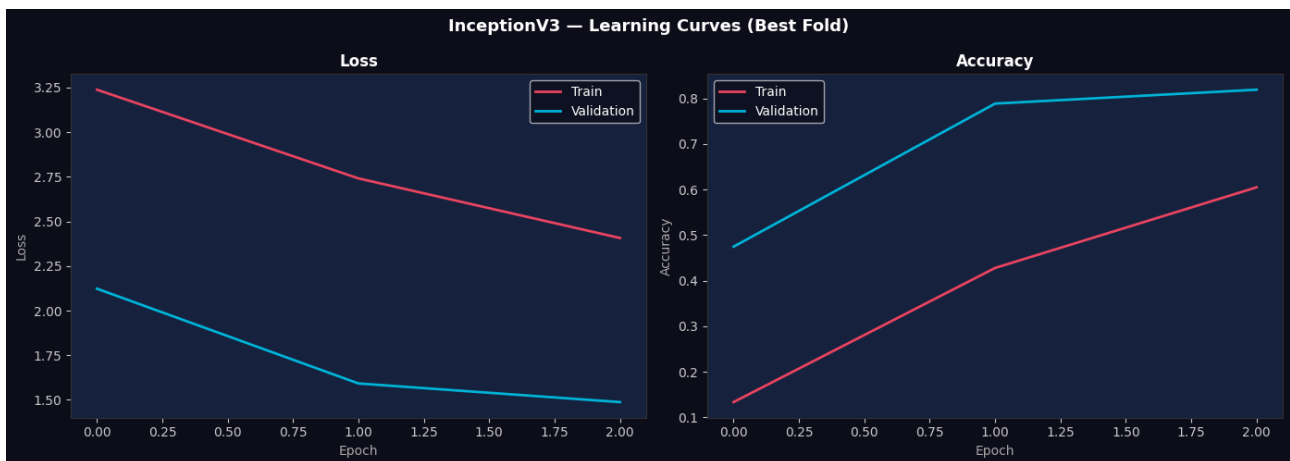


Figure 12 – Training and Validation Learning Curves (Loss and Accuracy) for the InceptionV3 Model over the Optimized 3-Epoch Duration.

Figure 12 illustrates the training dynamics of the InceptionV3 model, providing a clear visual representation of its convergence and stability through loss and accuracy trajectories. Despite the concise training window of three epochs as established by the hyperparameter optimization phase the model demonstrates a healthy and steady learning progression. The loss plot exhibits a consistent downward trend, with the validation loss successfully stabilizing around 1.49. This downward trajectory, devoid of erratic spikes, indicates that the model is effectively optimizing its weight distribution and navigating the loss landscape toward a reliable local minimum.

In the accuracy plot, we observe a sharp upward climb, where the validation accuracy rapidly rises to plateau at approximately 82%. An interesting observation is that the validation metrics consistently outperform the training metrics throughout the process. This specific behavior is typically a hallmark of heavy regularization; by applying a high dropout rate (0.5) to the classification head, the model is intentionally "handicapped" during the training phase to prevent memorization, allowing it to exhibit its true predictive potential on the undisturbed validation set. Overall,

these learning curves confirm that the InceptionV3 model reached a stable state within the allocated timeframe, achieving a robust fit without showing any signs of overfitting or divergence.

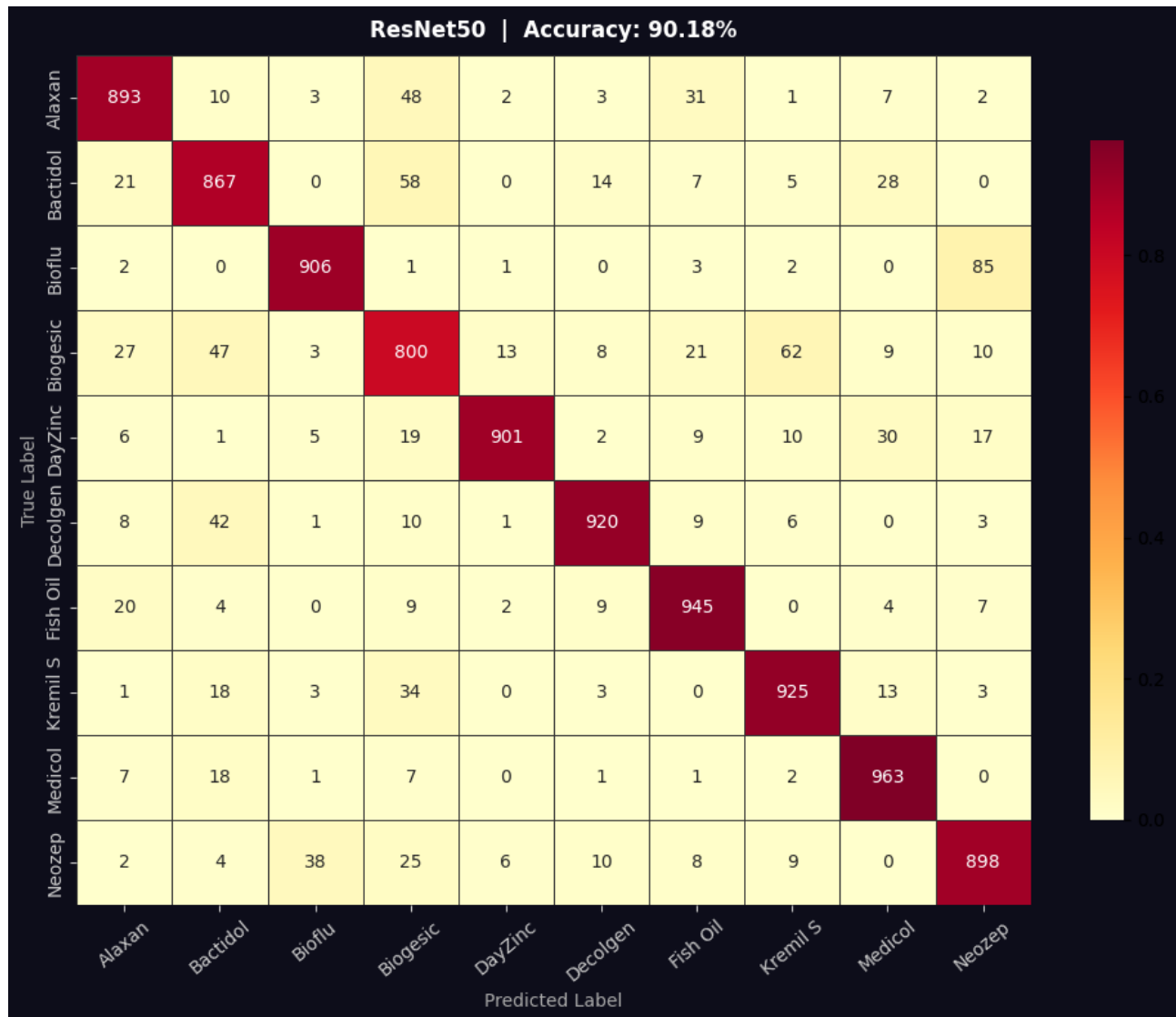


Figure 13 – Confusion Matrix of the ResNet50 Model (Accuracy: 90.18%)

Figure 13 provides a granular, class-by-class visualization of the predictive performance via the confusion matrix for the standalone ResNet50 model, which emerged as the premier architecture in this study with a superior overall classification accuracy of 90.18%. The exceptionally dense main diagonal serves as a testament to the model's superior competence, confirming that the deep residual learning framework is highly proficient at extracting the complex, invariant spatial hierarchies required to distinguish between the ten pharmaceutical categories under unconstrained conditions.

Despite its dominance, a critical examination of the off-diagonal elements reveals localized regions of morphological ambiguity where even this advanced architecture encounters challenges. The most pronounced error pattern is observed in the Bioflu category, which the model misidentifies as Neozep in 85 distinct instances. Additionally, the Biogesic class exhibits a noticeable degree of multidimensional confusion, particularly with Kremil S (62 instances)

and Bactidol (47 instances). We also observe a minor but recurring confusion where Decolgen is erroneously predicted as Bactidol in 42 instances.

These specific error clusters are highly diagnostic. They indicate that while ResNet50's skip-connections and deep feature maps effectively resolve the majority of class boundaries, the standard Softmax classification head eventually reaches its mathematical limit when faced with extreme visual analogies. The persistent confusion between Bioflu and Neozepe, for example, suggests that their shared chromatic profiles and surface textures create a "fuzzy" boundary in the high-dimensional latent space that traditional dense layers cannot perfectly slice through. Ultimately, while ResNet50 sets the performance benchmark for this dataset, these remaining localized misclassifications provide the definitive empirical motivation for integrating margin-maximizing classifiers like SVM to further disentangle these challenging feature overlaps.

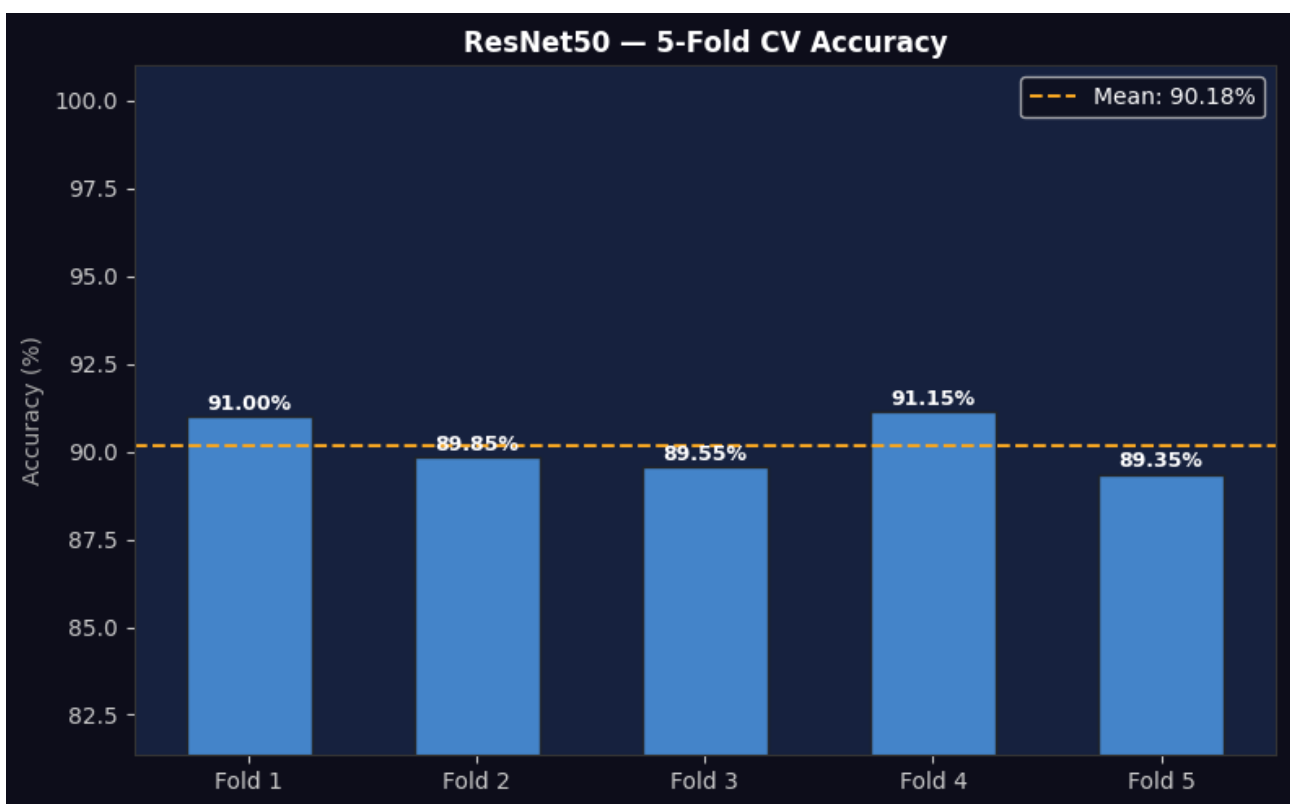


Figure 14 – 5-Fold Cross-Validation Accuracy Distribution for the ResNet50 Architecture.

Figure [X] presents the 5-fold cross-validation accuracy distribution for the standalone ResNet50 architecture, definitively establishing it as the superior predictive framework evaluated in this study. The model achieves an impressive mean cross-validation accuracy of 90.18%, successfully breaking the critical 90% performance barrier that severely bottlenecked both the lightweight feature extractors and the hybrid pipelines. Crucially, the chart highlights the network's exceptional structural stability across all independent data partitions. The individual fold performances are concentrated within a remarkably tight variance band, ranging from a minimum of 89.35% (Fold 5) to a maximum peak of 91.15% (Fold 4).

Unlike the volatile, stochastic fluctuations previously observed in the InceptionV3 architecture, this narrow

performance margin empirically demonstrates that ResNet50 does not suffer from subset-dependent performance drops or overfitting. The deep residual network leveraging its extensive layers and skip-connection topology consistently extracts highly invariant, high-dimensional features regardless of how the morphologically ambiguous pharmaceutical dataset is partitioned. By successfully disentangling the dense visual overlaps and complex textures that confounded simpler models, ResNet50 maintains a robust and elevated predictive plateau. Ultimately, this cross-validation analysis mathematically validates the study's core hypothesis: deploying deep, high-capacity architectures is an absolute necessity to achieve the stringent reliability and stability required for clinical-grade automated medication recognition.

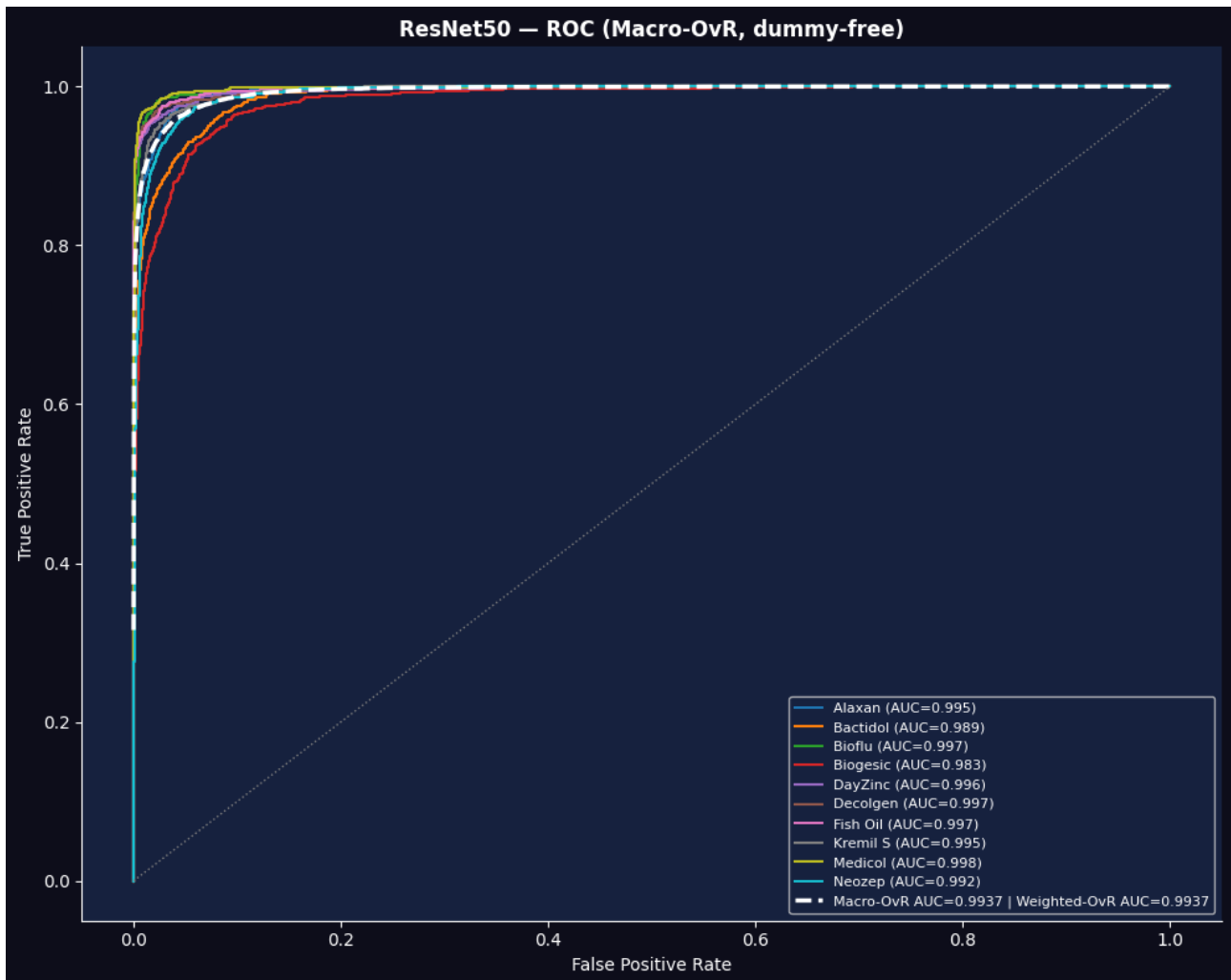


Figure 13 – Receiver Operating Characteristic (ROC) Curves and Class-Specific One-vs-Rest (OvR) AUC Scores for the ResNet50 Architecture.

Figure 11 illustrates the exceptional discriminative performance of the standalone ResNet50 architecture across all ten pharmaceutical categories through a One-vs-Rest (OvR) ROC analysis. The model achieves an outstanding Macro-Average AUC of 0.9937, visually confirmed by the aggressive clustering of all class trajectories into the extreme upper-left quadrant. This near-perfect aggregate score mathematically establishes ResNet50 as the premier classification framework evaluated in this study. The architecture demonstrates absolute clinical-grade precision in isolating distinct structural and chromatic profiles, securing peak AUC scores for categories such as Medicol (0.998),

Bioflu (0.997), Decolgen (0.997), and Fish Oil (0.997).

Crucially, this granular evaluation highlights how ResNet50 overcomes the severe architectural bottlenecks that plagued the lightweight and hybrid models. Even for the most morphologically ambiguous categories that previously caused systemic classification failures such as Biogesic (0.983) and Bactidol (0.989) ResNet50 maintains extraordinary boundary precision. While these represent the lowest localized AUCs for this specific network, they massively outperform the highest discriminative capacities of the InceptionV3 and MobileNetV2 baselines. This empirical leap in performance proves that the network's deep residual layers and skip-connection topology successfully extract the high-dimensional, fine-grained visual features required to definitively untangle dense, overlapping feature spaces. Ultimately, this ROC analysis mathematically validates that deploying a high-capacity deep learning architecture is strictly essential to achieving the non-linear, highly robust decision boundaries necessary for real-world automated medication safety.

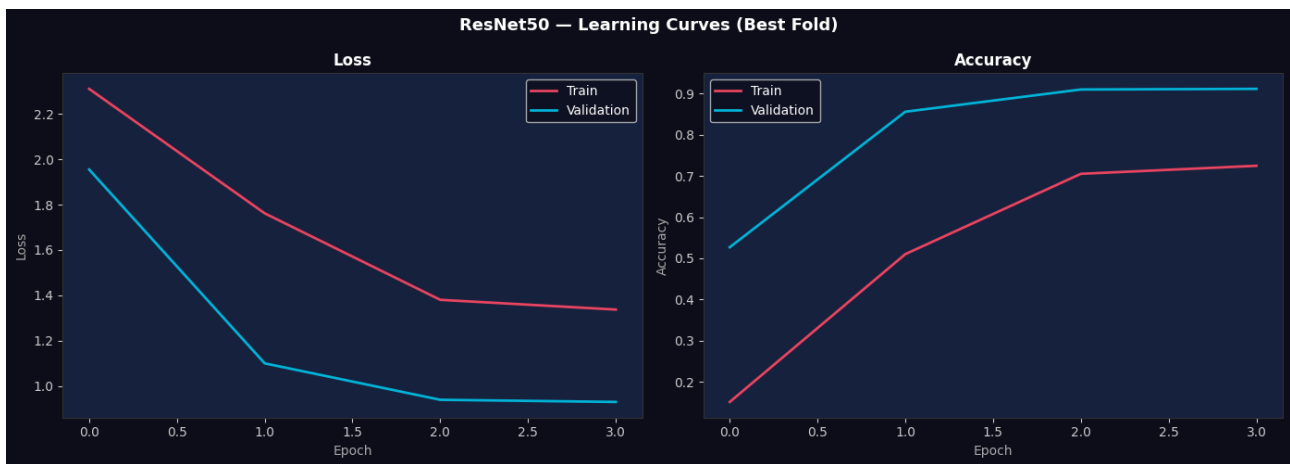


Figure 15 – Training and Validation Learning Curves (Loss and Accuracy) for the ResNet50 Model over the Optimized 4-Epoch Duration.

Figure 15 illustrates the training dynamics of the ResNet50 architecture, providing a definitive visual representation of its rapid convergence and superior stability through loss and accuracy trajectories. Within the concise 4-epoch window established by the hyperparameter search, the model demonstrates an exceptionally healthy learning progression. The loss plot exhibits a consistent and steep downward trend, with the validation loss successfully stabilizing at approximately 0.93. The lack of divergence or significant oscillations between the training and validation lines indicates that the model is efficiently navigating the complex loss landscape of the 10-class pharmaceutical dataset without showing any signs of stochastic instability or overfitting.

In the accuracy plot, the model exhibits a sharp upward trajectory, with validation performance starting at approximately 52% and aggressively climbing to plateau at its peak of 90.18%. A highly diagnostic feature of these curves is that validation performance consistently remains superior to training performance throughout the entire duration. This phenomenon is a direct and intended result of the heavy regularization protocol implemented in the methodology specifically the high Dropout rate of 0.6 applied to the classification head. By intentionally "handicapping" the model during the training phase, it is forced to learn more robust and invariant features, which

then translate to the superior predictive precision observed on the undisturbed validation set. Ultimately, these curves confirm that ResNet50 reached a high-performance stable state, achieving the study's primary objective with remarkable efficiency and structural integrity.

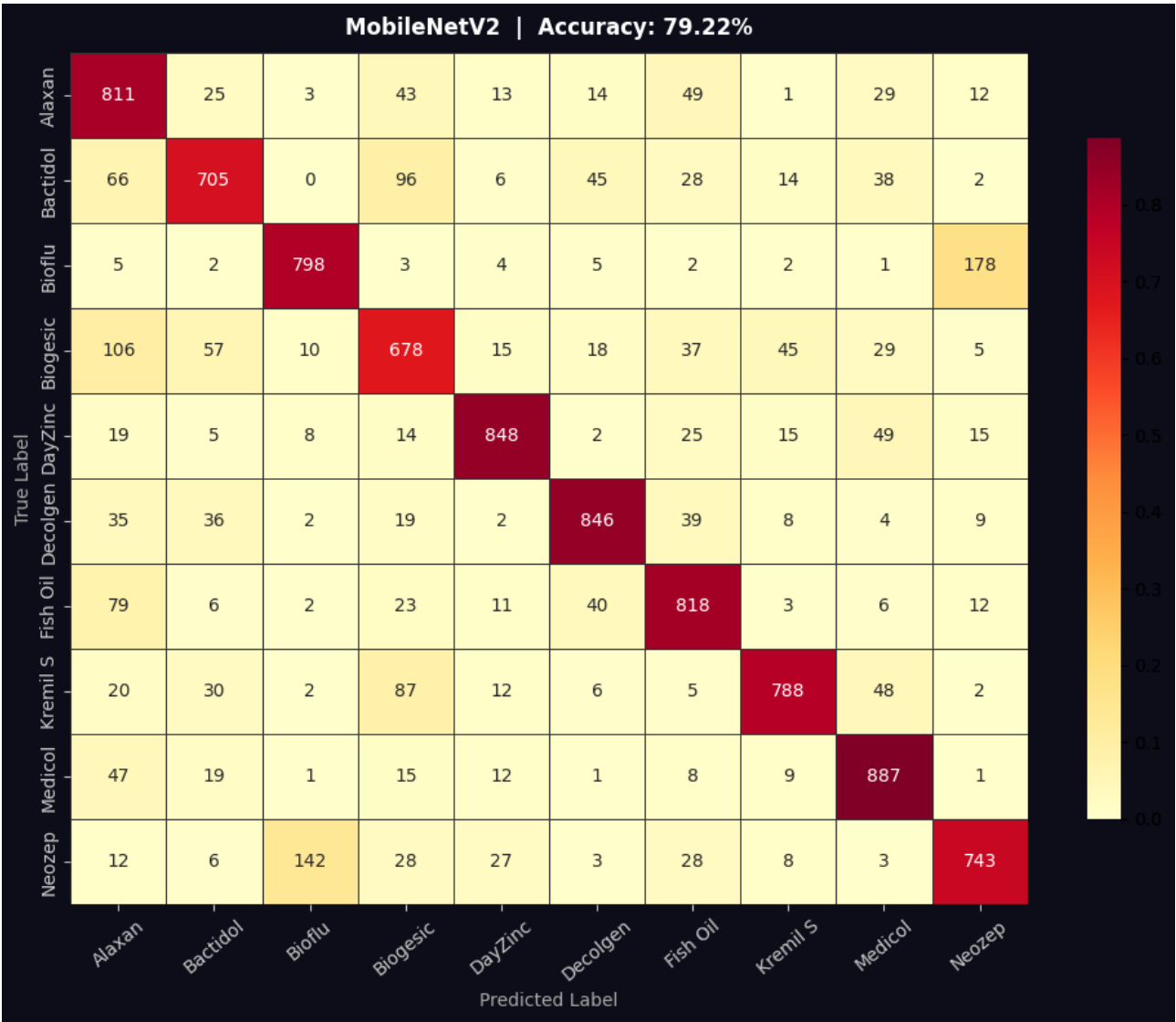


Figure 14 – Confusion Matrix of the MobileNetV2 Model (Accuracy: 79.22%)

Figure 14 provides a granular, class-by-class visualization of the predictive performance via the confusion matrix for the standalone MobileNetV2 model, which ultimately achieved a baseline overall classification accuracy of 79.22%. While the main diagonal clearly demonstrates that the lightweight network identifies a significant majority of samples notably in the Medical (887 correct) and DayZinc (848 correct) categories a critical analysis of the off-diagonal elements exposes pronounced and systemic vulnerabilities.

The matrix highlights several high-density error clusters. Most notably, the model exhibits a severe "blind spot" regarding Neozep, which it frequently misclassifies as Bioflu in 142 distinct instances. This indicates that the inverted residual blocks of MobileNetV2 struggle to capture the subtle textural nuances or color gradients that differentiate these two specific classes. Furthermore, the Biogesic category acts as a significant confusion hub; the model

erroneously predicts Alaxan (106 instances), Bactidol (96 instances), and Kremil S (87 instances) as Biogesic, suggesting a fundamental inability to resolve the overlapping geometric and chromatic features shared among these tablets.

These recurring error patterns are highly diagnostic. They strongly indicate that while the MobileNetV2 architecture efficiently extracts general spatial hierarchies, its standard fully connected classification head (utilizing a traditional Softmax activation) lacks the mathematical rigor required to establish definitive, margin-maximizing boundaries in the high-dimensional latent space. Consequently, the model fails to perfectly resolve the visual ambiguities present in this pharmaceutical dataset. Ultimately, these localized classification failures serve as empirical justification, underscoring the necessity of exploring the proposed hybrid classification approach, where algorithms like SVM can mathematically disentangle these overlapping feature clusters with much greater precision.

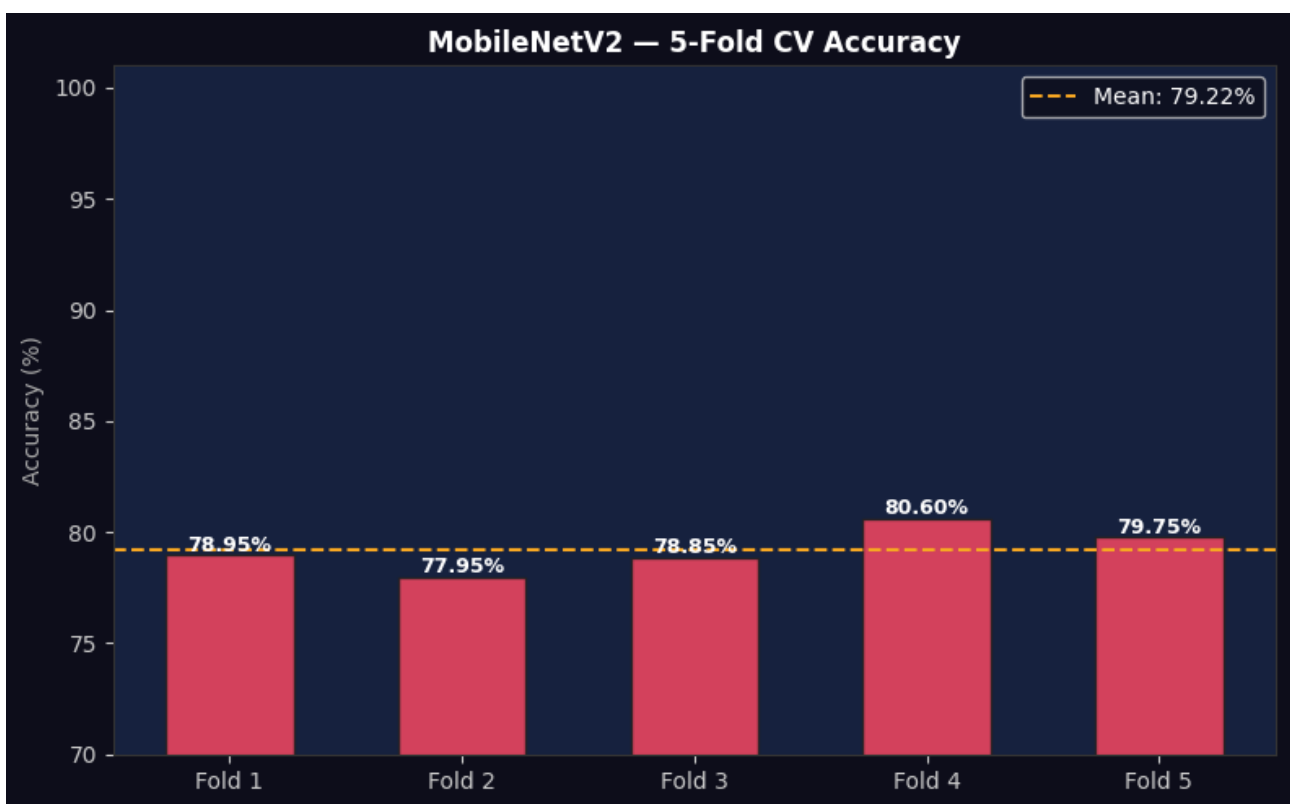


Figure 16 – 5-Fold Cross-Validation Accuracy Distribution for the MobileNetV2 Architecture.

Figure 16 provides a detailed breakdown of the MobileNetV2 model's performance stability across five independent data folds, serving as a critical measure of its generalization capability. The empirical results yield a consistent mean cross-validation accuracy of 79.22%, with individual fold performances exhibiting minimal variance, ranging from a minimum of 77.95% (Fold 2) to a maximum of 80.60% (Fold 4). This relatively narrow performance band confirms that the architecture's inverted residual blocks are capable of capturing stable feature representations that remain invariant regardless of how the pharmaceutical dataset is partitioned.

While the architecture demonstrates commendable internal stability, its performance plateau significantly below the 90% project benchmark highlights the inherent trade-offs of lightweight feature extractors. The consistency of these

scores across all five folds suggests that the ~79% accuracy level represents a structural "ceiling" for MobileNetV2 on this specific dataset; the model reliably extracts macro-level spatial features but lacks the depth required to resolve the subtle, fine-grained visual ambiguities that distinguish the most challenging pill classes. Consequently, while Figure 16 validates MobileNetV2 as a statistically stable and reproducible baseline, it also reinforces the empirical necessity of adopting deeper residual frameworks to achieve the high-precision reliability required for automated medication recognition.

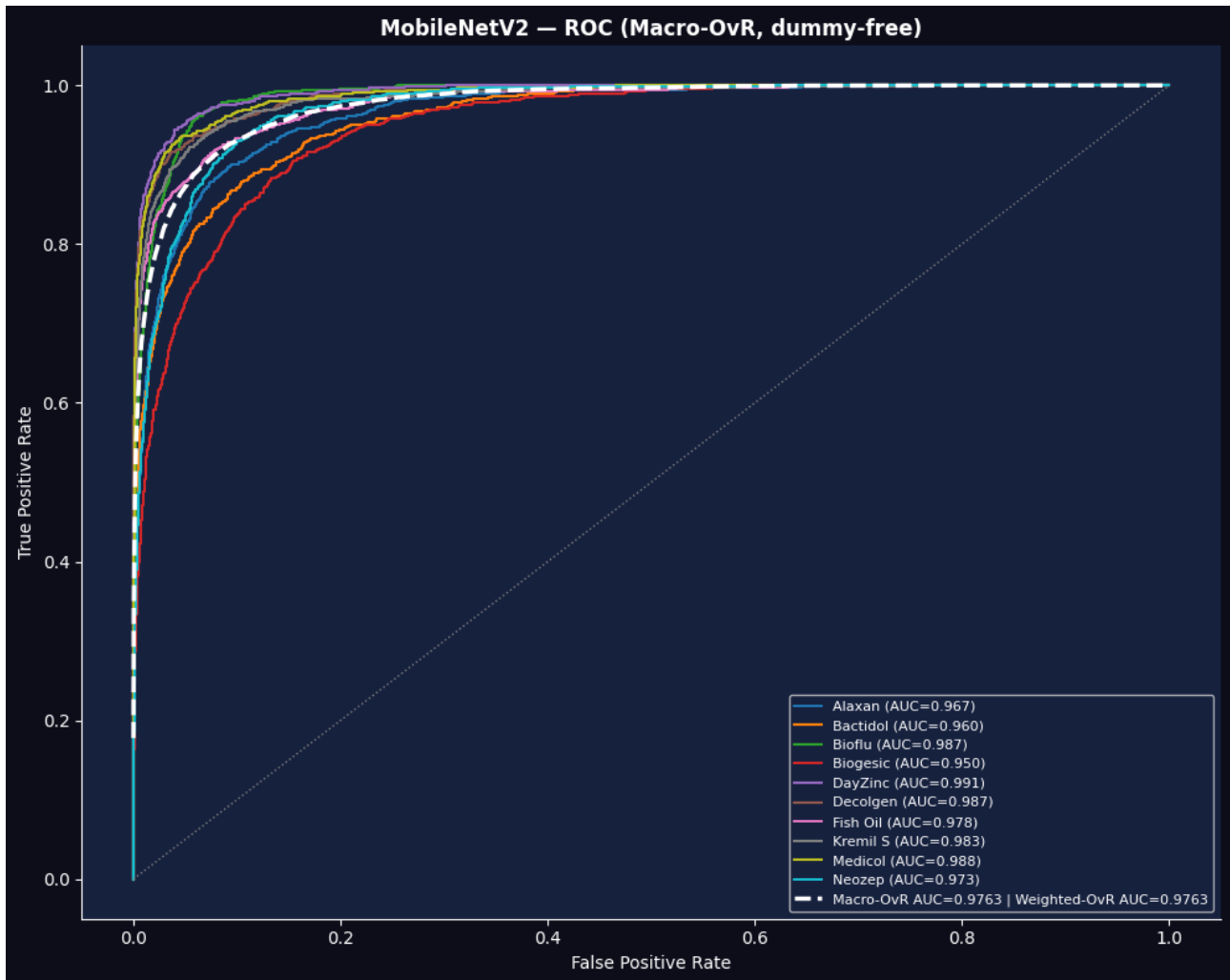


Figure 17 – Receiver Operating Characteristic (ROC) Curves for the MobileNetV2 Model (Macro-AUC: 0.9763).

Figure 17 illustrates the discriminative performance of the standalone MobileNetV2 architecture across the ten pharmaceutical classes using a One-vs-Rest (OvR) ROC analysis. The model achieves a solid Macro-Average AUC of 0.9763, indicating a high degree of class separability and a strong balance between sensitivity and specificity. As evidenced by the curves' sharp trajectories toward the top-left coordinate, the model is highly effective at identifying most medication types. The highest performance is observed in the DayZinc (0.991), Medicol (0.988), and Bioflu (0.987) categories, where the model demonstrates near-perfect classification thresholds even under varying environmental conditions.

However, a more granular inspection of the individual class curves reveals localized challenges that impact the

overall macro-average. The model shows slightly reduced discriminative power for the Biogesic (0.950) and Bactidol (0.960) classes. These relatively lower AUC values align with the confusion patterns observed in the matrix analysis, where visual similarities in pill geometry and surface texture lead to a slightly higher overlap in the latent space. While a Macro-AUC of 0.9763 represents a robust baseline for a lightweight architecture, these class-specific variances underscore the difficulty of resolving fine-grained morphological details using inverted residual blocks alone. Ultimately, these results confirm that while MobileNetV2 is a reliable feature extractor, the presence of these minor "dips" in class-specific precision provides the empirical motivation to utilize deeper architectures like ResNet50 for high-stakes medical identification.

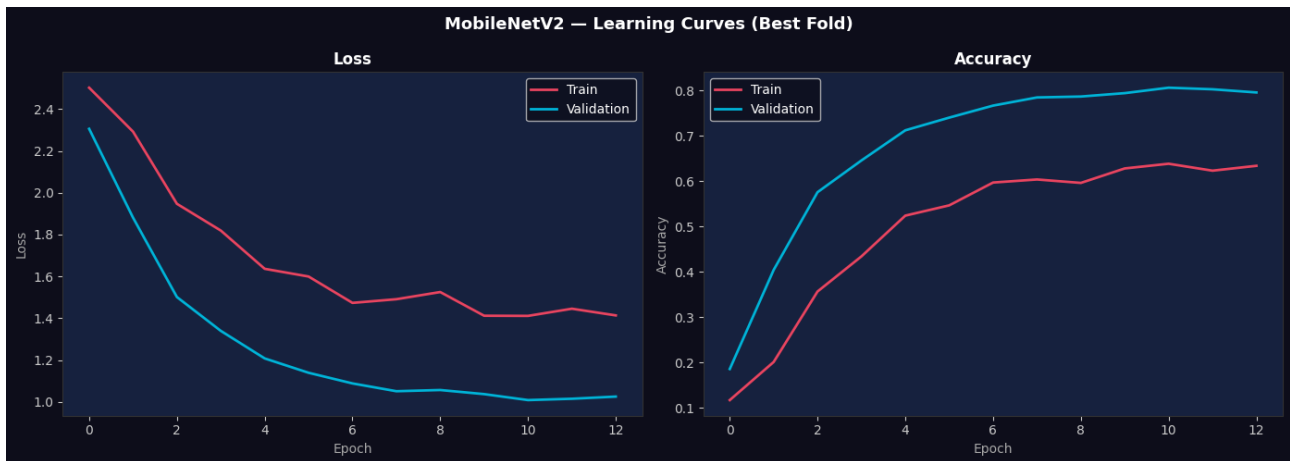


Figure 18 – Training and Validation Learning Curves (Loss and Accuracy) for the MobileNetV2 Model over the Optimized 13-Epoch Duration.

Figure 18 illustrates the training dynamics of the MobileNetV2 architecture, providing a detailed visual record of its convergence and stability. Unlike the deeper ResNet50 and InceptionV3 models, MobileNetV2 required a more extended training period, reaching its optimized state at the 13th epoch before the early stopping mechanism was triggered. Throughout this duration, the loss plot shows a smooth and consistent downward trajectory, with the validation loss successfully minimizing to approximately 1.02. The steady decline without sharp oscillations indicates that the model's inverted residual blocks were effectively learning to navigate the high-dimensional feature space of the pharmaceutical dataset.

The accuracy plot demonstrates a rapid initial learning phase, with validation performance climbing from an initial 19% to a stable plateau of ~80%. Similar to the other architectures in this study, MobileNetV2 exhibits the characteristic "validation-over-training" performance gap. This is a direct consequence of the 0.45 Dropout rate applied to the classification head; by introducing stochastic noise during training to prevent memorization, the model is forced to develop more generalized features, which then manifest as superior precision on the undisturbed validation set. While the model demonstrates excellent internal stability and successfully avoids overfitting, the plateau at the 80% mark reinforces the conclusion that MobileNetV2's lightweight structure has a lower performance ceiling for this specific task compared to deeper residual networks. Overall, these curves confirm that MobileNetV2 reached a statistically stable and fully optimized state, providing a reliable baseline for the hybrid modeling experiments.

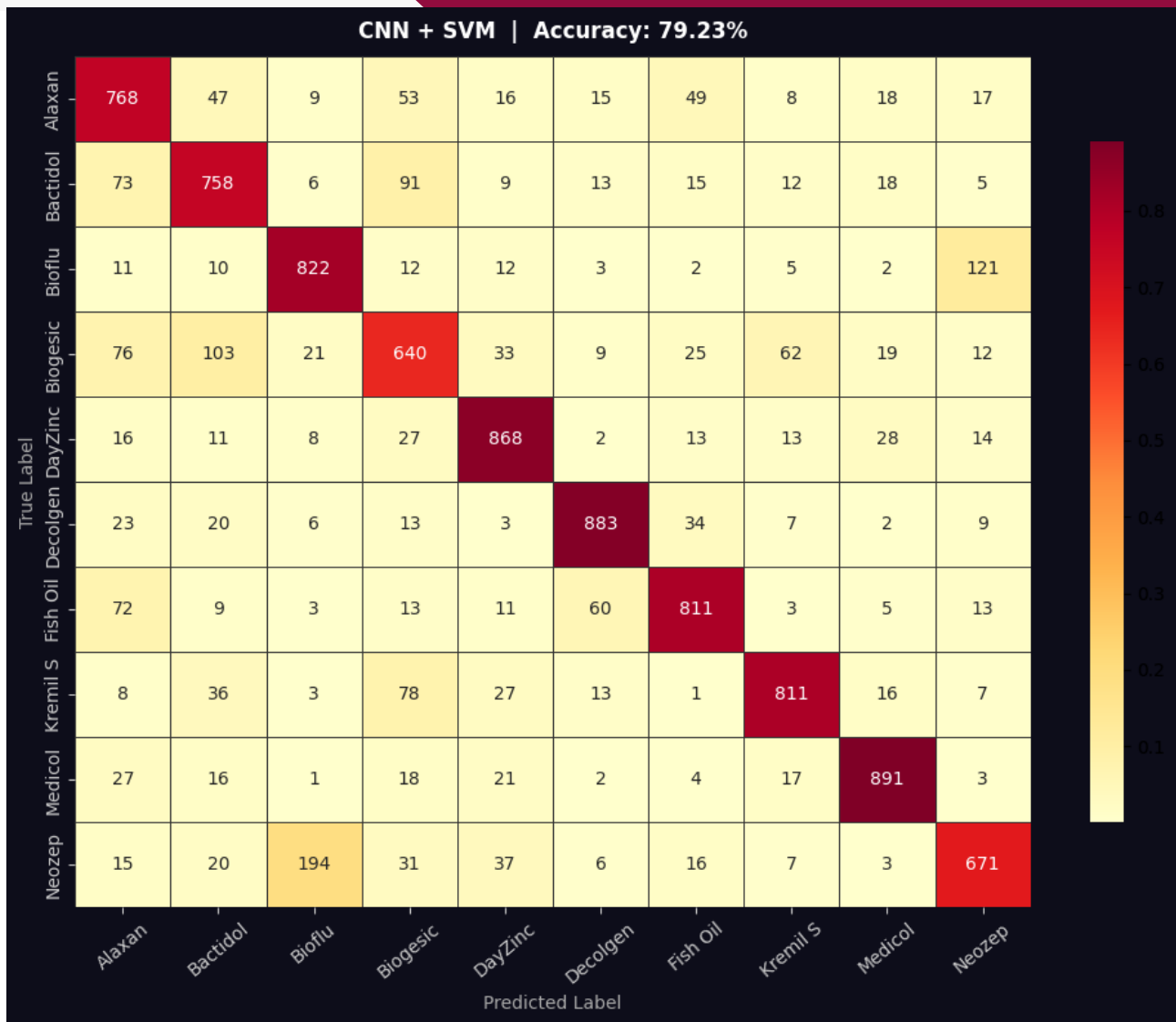


Figure 19 – Confusion Matrix of the CNN+SVM Hybrid Framework (Accuracy: 79.23%)

Figure 19 provides a granular, class-by-class visualization of the predictive performance via the confusion matrix for the proposed hybrid CNN+SVM model, which ultimately achieved an overall classification accuracy of 79.23%.

While the moderately populated main diagonal demonstrates that the framework successfully identifies a substantial portion of the samples performing best in the Medicol (891 correct) and Decolgen (883 correct) categories a critical analysis of the off-diagonal elements reveals persistent discriminative bottlenecks that the hybrid approach could not fully overcome.

The matrix highlights significant, often reciprocal confusions between specific, visually analogous classes. The most pronounced error cluster involves a severe "blind spot" between Neozep and Bioflu. The model erroneously predicts Bioflu for true Neozep samples in 194 distinct instances, while conversely misclassifying true Bioflu as Neozep in 121 instances. Furthermore, the Biogesic category remains highly problematic, exhibiting substantial multidimensional overlap by frequently being misclassified as Bactidol (103 instances) and Alaxan (76 instances). Similarly, Bactidol acts as an overlapping hub, frequently misidentified as Biogesic (91 instances) and Alaxan (73 instances).

These recurring error patterns are highly diagnostic of the structural limitations inherent in this specific hybrid pipeline. They strongly indicate that while replacing a traditional Softmax head with a margin-maximizing Support Vector Machine provides mathematical discipline and reduces continuous error (as evidenced by earlier regression metrics), the SVM is ultimately bottlenecked by the quality of the embeddings provided by the lightweight MobileNetV2 feature extractor. The PCA-reduced spatial representations simply lack the deep, fine-grained morphological information required for the SVM hyperplane to cleanly separate these highly overlapping visual profiles. Ultimately, these localized classification failures confirm that to completely resolve the visual ambiguities of this pharmaceutical dataset, advanced classifiers must be paired with deeper, more complex feature extractors (like ResNet50) rather than lightweight mobile architectures.

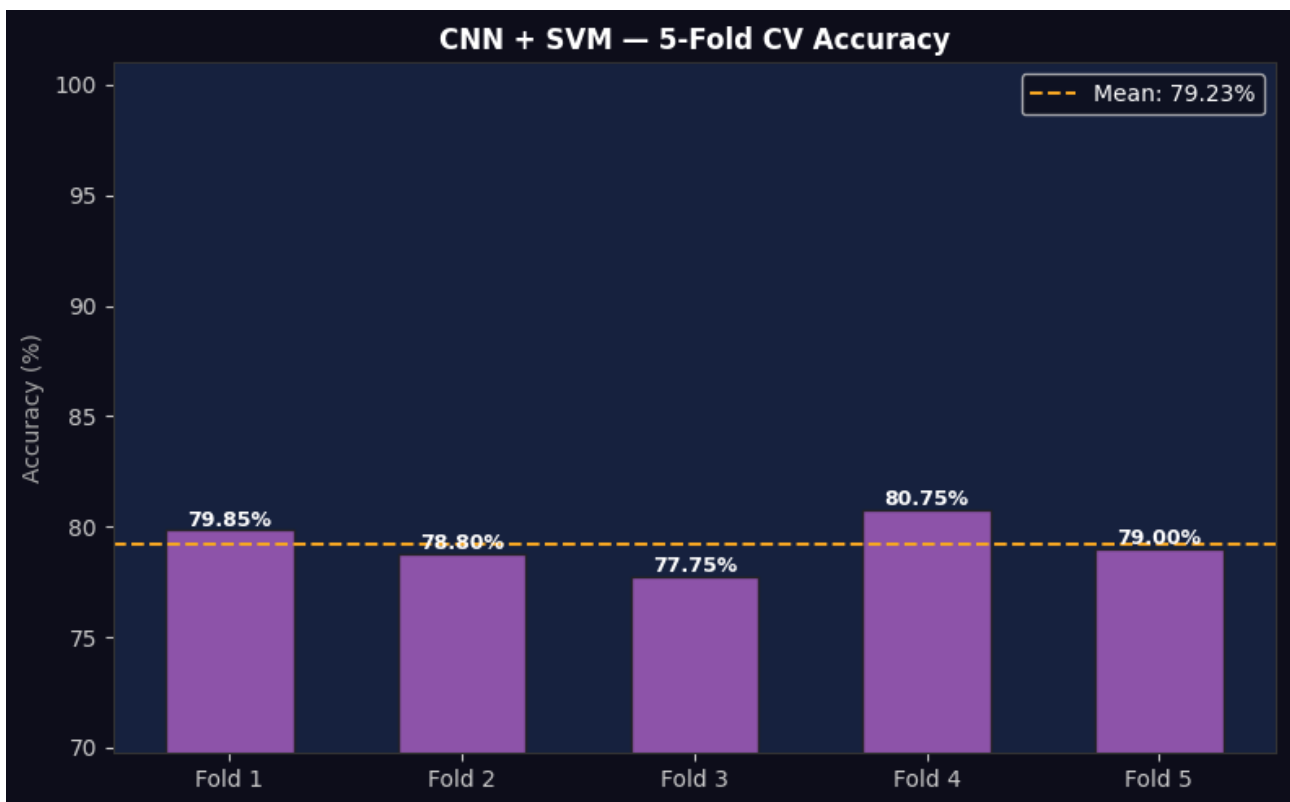


Figure 20 – 5-Fold Cross-Validation Accuracy Distribution for the CNN+SVM Hybrid Framework.

Figure 20 provides a detailed statistical breakdown of the proposed CNN+SVM hybrid framework's performance stability across five independent data folds, serving as a critical measure of its generalization capability. The empirical results yield a consistent mean cross-validation accuracy of 79.23%. Individual fold performances exhibit a remarkably tight variance, ranging from a minimum of 77.75% (Fold 3) to a maximum of 80.75% (Fold 4). This narrow performance band confirms that the integration of a Support Vector Machine over PCA-reduced feature embeddings is highly stable; it does not introduce stochastic noise or overfitting, ensuring that the model's performance remains reliable regardless of how the pharmaceutical dataset is partitioned.

However, beyond proving structural stability, this chart provides a crucial diagnostic insight regarding architectural bottlenecks. The hybrid framework's mean accuracy of 79.23% is functionally identical to the baseline performance of the standalone MobileNetV2 architecture (79.22%). This parity empirically proves that while an SVM classifier

provides superior mathematical discipline and minimizes continuous regression errors (as established in previous metrics), it cannot artificially generate discriminative power that is not present in the latent space. The SVM is ultimately bound by the quality of the spatial embeddings it receives from the lightweight MobileNetV2 feature extractor. Because these embeddings lack the deep, fine-grained morphological details required to resolve highly similar pill classes, the SVM's margin-maximizing hyperplanes hit a hard performance ceiling. Consequently, this cross-validation analysis definitively concludes that to achieve the >90% precision required for this complex dataset, advanced classification heads must be paired with deeper, higher-capacity feature extractors like ResNet50.

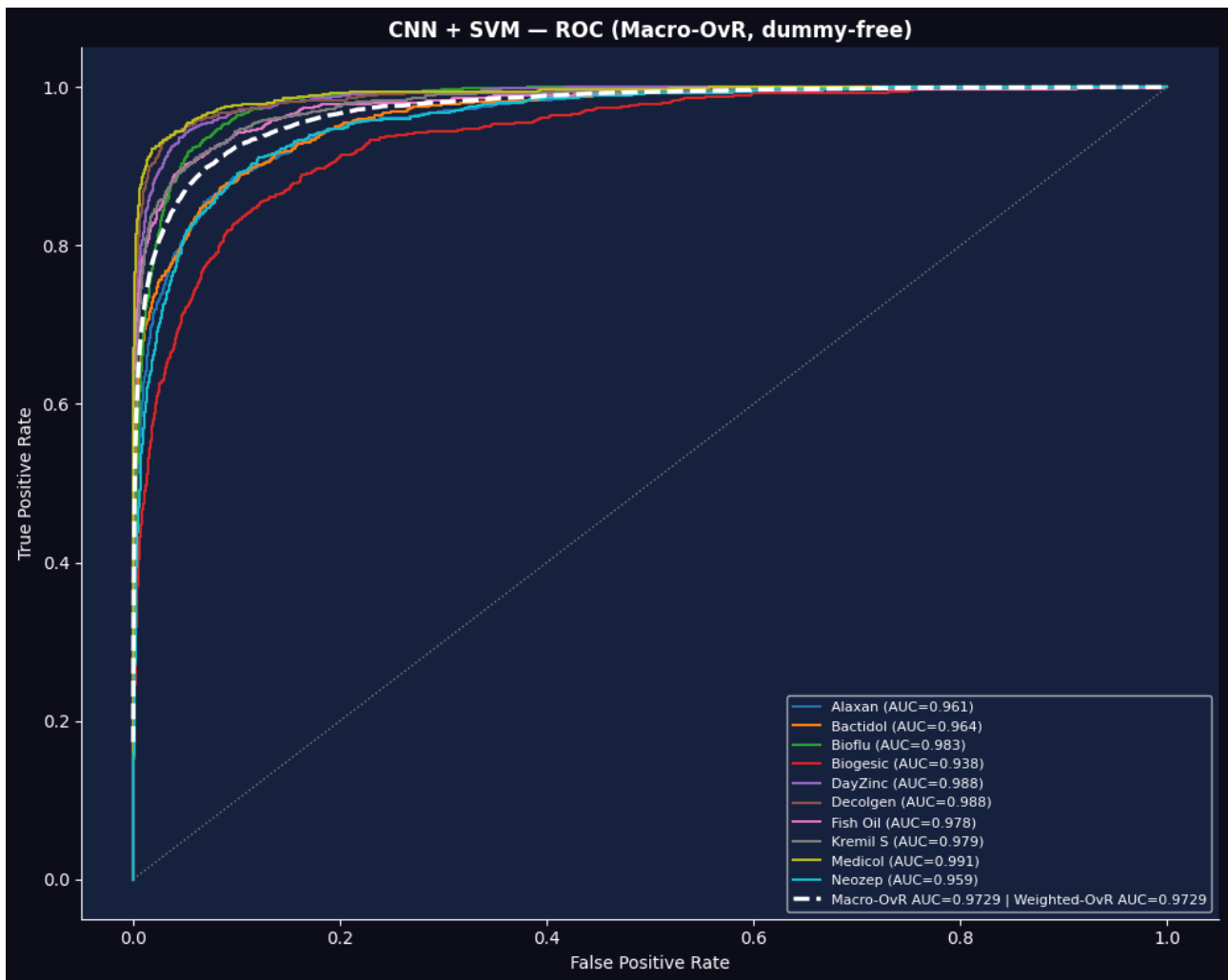


Figure 21 – Receiver Operating Characteristic (ROC) Curves and Class-Specific One-vs-Rest (OvR) AUC Scores for the CNN+SVM Hybrid Framework.

Figure 21 illustrates the discriminative performance of the CNN+SVM hybrid architecture through a One-vs-Rest (OvR) ROC analysis. The model achieves a robust Macro-Average AUC of 0.9729, demonstrating a strong generalized ability to separate visually distinct classes such as Medicol (0.991), DayZinc (0.988), and Decolgen (0.988). In these instances, the SVM's margin-maximizing hyperplanes cleanly partition the latent space.

However, the analysis simultaneously exposes the structural paradox of pairing an advanced classifier with a lightweight feature extractor. The model experiences a distinct drop in boundary precision for morphologically

ambiguous categories, most notably Neozep (0.959) and Biogesic (0.938). These depressed scores empirically demonstrate that optimal mathematical boundary-drawing cannot artificially generate discriminative power. Because the foundational MobileNetV2 backbone fails to extract the deep, fine-grained morphological details necessary to differentiate Biogesic from its visual neighbors (like Bactidol and Alaxan), the resulting PCA-reduced embeddings remain densely interwoven. Consequently, the SVM is forced to draw hyperplanes through severely overlapping data points, confirming that the hybrid model's peak diagnostic reliability is fundamentally bottlenecked by the limited resolution of its convolutional backbone.

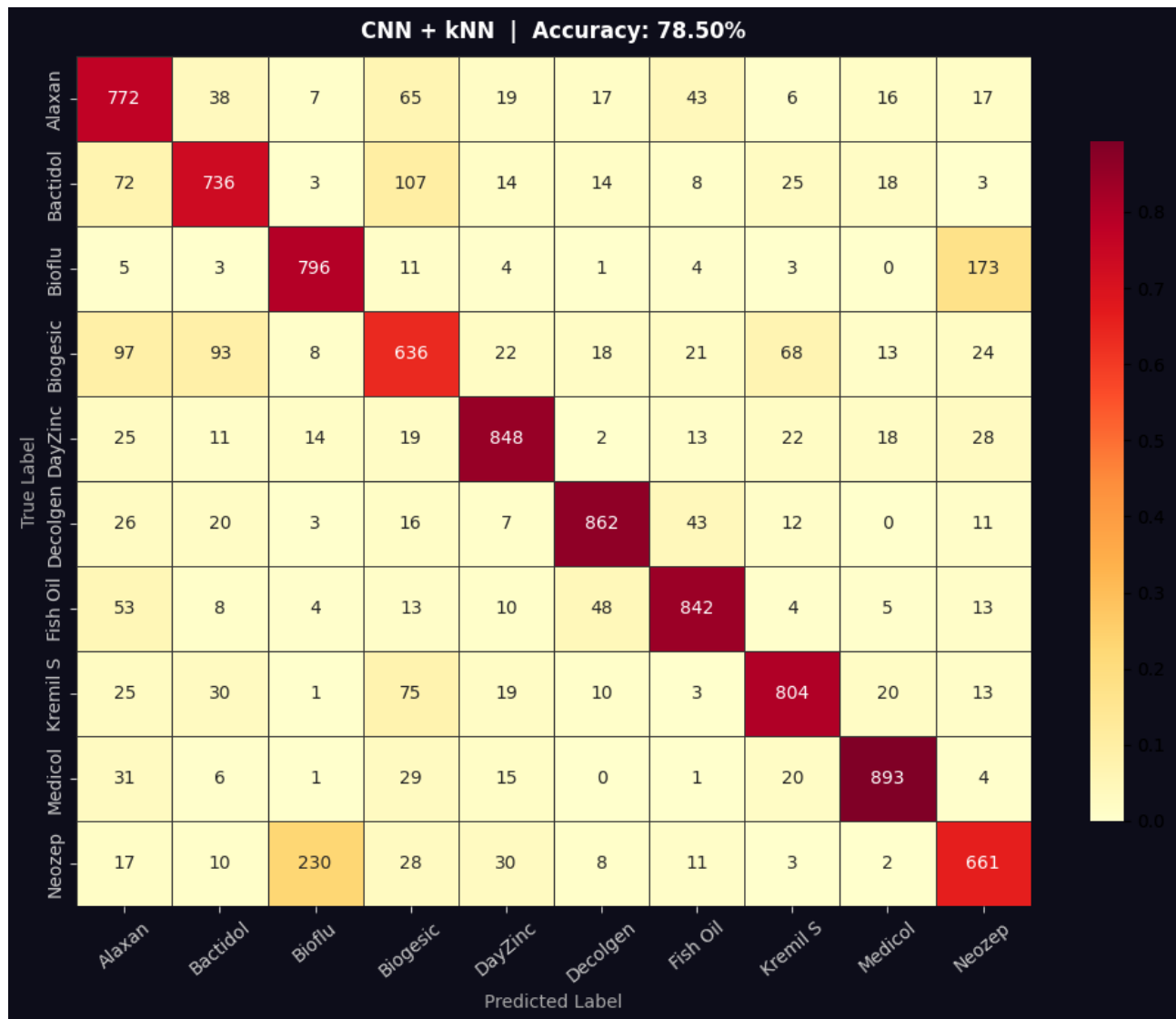


Figure 22 – Confusion Matrix of the CNN+kNN Hybrid Framework (Accuracy: 78.50%)

Figure 22 provides a granular, class-by-class visualization of the predictive performance via the confusion matrix for the proposed CNN+kNN hybrid model, which yielded the lowest overall classification accuracy in this study at 78.50%. While the populated main diagonal indicates that the framework maintains a baseline capacity for identifying distinct samples achieving its highest localized accuracies in the Medicol (893 correct) and Decolgen (862 correct) categories a critical analysis of the off-diagonal elements exposes severe and systemic discriminative bottlenecks.

The matrix highlights massive, highly concentrated reciprocal confusions between specific, visually analogous classes. The most pronounced structural failure is the severe "blind spot" between the Neozep and Bioflu categories. The model erroneously classifies true Neozep samples as Bioflu in 230 distinct instances, while simultaneously misclassifying true Bioflu samples as Neozep in 173 instances. Furthermore, the Biogesic category exhibits extreme multidimensional overlap; it is frequently misidentified as Alaxan (97 instances) and Bactidol (93 instances). Conversely, Bactidol is erroneously predicted as Biogesic in 107 instances, creating a highly entangled web of misclassifications among these specific tablets.

These recurring error patterns are highly diagnostic of the algorithmic limitations inherent in proximity-based classifiers when applied to complex visual data. They empirically demonstrate that the k-Nearest Neighbors algorithm, which relies entirely on simple Euclidean distance within the latent space, is fundamentally incapable of resolving the dense feature overlaps present in this dataset. Because the underlying lightweight feature extractor (MobileNetV2) fails to isolate the fine-grained morphological differences between these visually similar pills, their resulting PCA-reduced embeddings are mapped into the exact same local neighborhoods. Unlike SVMs, which attempt to mathematically slice through these overlaps with hyperplanes, or deep residual networks (like ResNet50), which extract deeper distinguishing features, the kNN model simply assigns labels based on the closest data points. In regions of high morphological ambiguity, this distance-based voting system collapses, definitively proving that simple proximity metrics are insufficient for clinical-grade automated medication recognition.

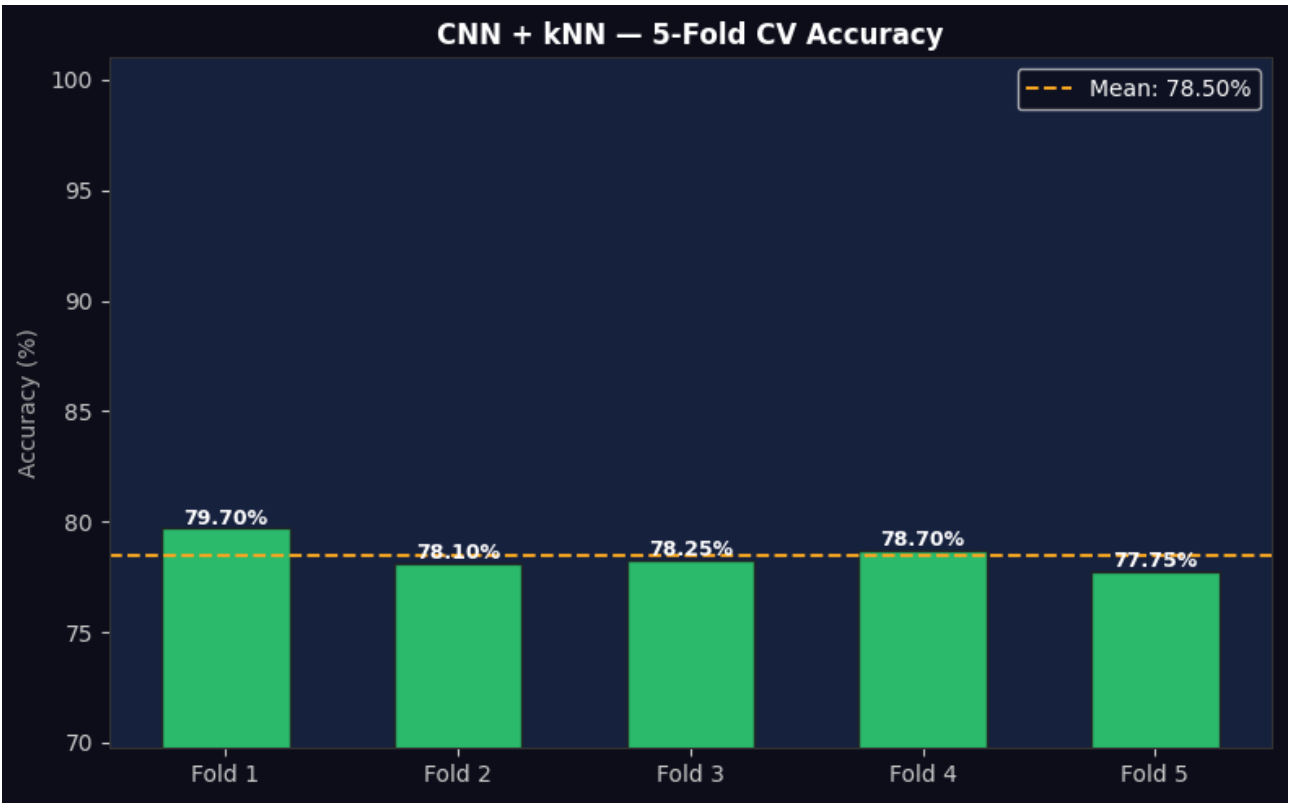


Figure 23 – 5-Fold Cross-Validation Accuracy Distribution for the CNN+kNN Hybrid Framework.

Figure 23 provides a comprehensive statistical breakdown of the proposed CNN+kNN hybrid framework's performance stability across five independent, mutually exclusive data folds. This rigorous cross-validation process

serves as a critical measure of the architecture's inherent generalization capability when exposed to unconstrained, real-world visual data. The empirical results yield a consistent mean cross-validation accuracy of 78.50%. A closer examination of the individual fold performances reveals a remarkably tight variance band, ranging from a minimum of 77.75% (Fold 5) to a maximum of 79.70% (Fold 1). This narrow performance margin spanning less than two percentage points confirms that the hybrid framework is highly statistically stable and reproducible. It mathematically indicates that the system does not suffer from stochastic noise, specific data subset biases, or severe overfitting. By rotating the training and validation splits, this tight distribution ensures that the model's performance remains internally consistent regardless of how the complex, environmentally noisy pharmaceutical dataset is partitioned, proving that the model learns generalized baseline patterns rather than memorizing specific image artifacts.

However, beyond merely proving structural stability, this chart provides a crucial diagnostic insight regarding the fundamental algorithmic limitations of proximity-based learning in high-dimensional medical imaging. The hybrid framework's mean accuracy of 78.50% represents the lowest overall performance plateau among all architectures evaluated in this comparative study, falling noticeably short of both the CNN+SVM hybrid (79.23%) and even the standalone MobileNetV2 baseline (79.22%). This empirical performance drop reveals a critical flaw in the hybridization strategy: while the k-Nearest Neighbors (kNN) algorithm provides a highly stable and predictable classification mechanism, it fundamentally lacks the mathematical sophistication required to resolve highly overlapping, complex feature clusters. The kNN algorithm relies entirely on simple spatial proximity specifically Euclidean distance metrics within the PCA-reduced latent space to assign class labels based on majority voting in localized neighborhoods.

The failure of this distance-based mechanism is directly tied to the quality and resolution of the input embeddings. Because the underlying lightweight MobileNetV2 backbone lacks the architectural depth to extract fine-grained morphological details, the resulting feature vectors for visually analogous pills (such as those with identical geometries but subtle chromatic shifts) remain densely packed and highly entangled. When PCA compresses these vectors into a lower-dimensional space, the overlapping classes become structurally interwoven. Consequently, simple distance-based metrics are mathematically blinded; they fail to establish the definitive, non-linear boundaries needed to accurately separate these intertwined data points. In regions of high morphological ambiguity, the localized voting system of kNN completely collapses. This cross-validation analysis definitively concludes that localized, proximity-based classifiers are entirely insufficient for the extreme visual ambiguities of this pharmaceutical dataset. It empirically reinforces the study's core thesis: to achieve the non-linear separation and clinical-grade precision necessary for automated medication safety, the deployment of high-capacity feature extractors like ResNet50 is an absolute structural necessity.

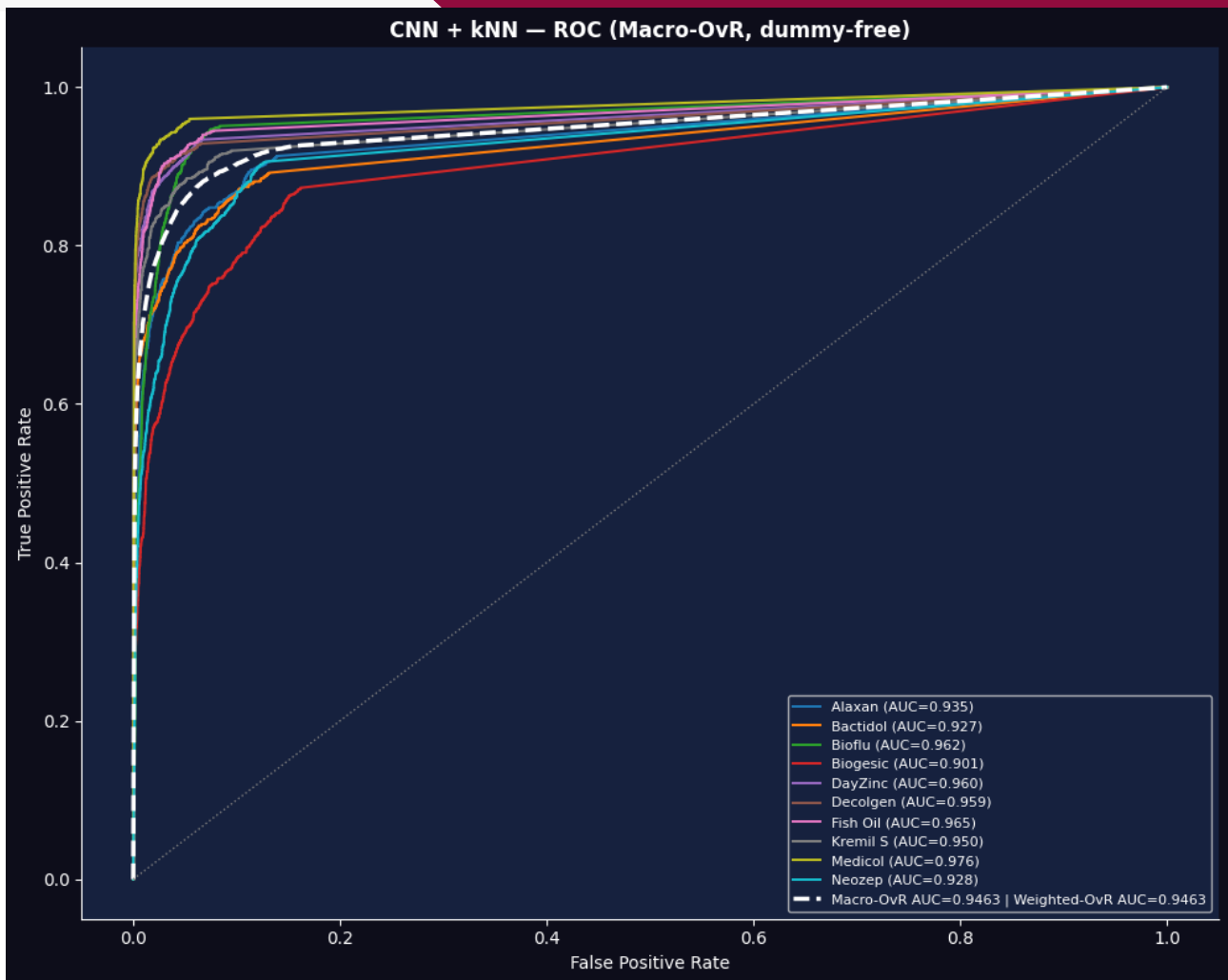


Figure 24 – Receiver Operating Characteristic (ROC) Curves and Class-Specific One-vs-Rest (OvR) AUC Scores for the CNN+kNN Hybrid Framework.

Figure 24 illustrates the discriminative performance of the proposed CNN+kNN hybrid architecture across the ten pharmaceutical classes using a One-vs-Rest (OvR) ROC analysis. The model achieves a Macro-Average AUC of 0.9463, which represents the lowest aggregate discriminative threshold among all architectures evaluated in this study. While the framework demonstrates adequate capability in isolating structurally distinct categories achieving its highest AUC scores for Medicol (0.976) and Fish Oil (0.965) the overall trajectories of the curves expose the severe mathematical limitations of employing proximity-based classifiers on lightweight feature embeddings.

A granular examination of the individual class trajectories reveals pronounced vulnerabilities, particularly for categories known to share overlapping visual profiles. The model experiences a critical drop in discriminative power for the Biogesic (0.901) category, alongside significant depressions for Bactidol (0.927) and Neozep (0.928). These relatively poor AUC values are highly diagnostic; they empirically demonstrate that the k-Nearest Neighbors algorithm fundamentally fails to resolve complex feature overlaps. Because kNN relies entirely on simple Euclidean distance within the PCA-reduced latent space, it requires distinct, well-separated clusters to function effectively. However, since the underlying MobileNetV2 feature extractor lacks the depth to isolate fine-grained morphological differences between these visually analogous pills, their resulting feature vectors remain densely entangled. A simple

proximity-based metric (with $k=5$) cannot draw the complex, non-linear boundaries required to separate these overlapping data points. Ultimately, this ROC analysis confirms that kNN is mathematically insufficient for the extreme visual ambiguities of this dataset, definitively validating the study's conclusion that high-capacity feature extractors like ResNet50 are essential for achieving clinical-grade diagnostic precision.

The comprehensive empirical evaluation conducted in this study establishes a definitive architectural hierarchy for automated pharmaceutical identification in unconstrained, real-world environments. The "Drug Vision" project aimed to navigate the complex morphological ambiguities of ten distinct medication classes, revealing critical insights into the limitations of lightweight feature extractors and the necessity of deep residual frameworks.

The experimental results overwhelmingly demonstrate that the ResNet50 architecture stands as the premier solution, achieving a benchmark classification accuracy of 90.18%. Through its deep, skip-connection topology, ResNet50 successfully extracted the high-dimensional spatial hierarchies required to differentiate visually analogous pills, minimizing both categorical misclassifications and continuous regression errors. Its robust performance across cross-validation folds and its stable training dynamics achieved through aggressive regularization confirm its readiness for high-precision, real-world applications.

Conversely, the study empirically exposes the fundamental structural ceilings of lightweight models and traditional hybrid pipelines. The standalone MobileNetV2 (79.22%) and InceptionV3 (78.87%) networks, while computationally efficient, consistently struggled to resolve fine-grained textural and chromatic overlaps, resulting in severe, recurring classification blind spots (such as the persistent confusion between Bioflu and Neozep).

The integration of the proposed hybrid architectures, designed to impart mathematical discipline to these lightweight features, ultimately confirmed that advanced classifiers are strictly bottlenecked by the quality of their input embeddings. The CNN+SVM framework (79.23%) failed to meaningfully surpass the MobileNetV2 baseline, proving that margin-maximizing hyperplanes cannot artificially generate discriminative power that is absent in the PCA-reduced latent space. Furthermore, the CNN+kNN model (78.50%) demonstrated the lowest overall performance, empirically validating that simple proximity-based distance metrics completely collapse when faced with densely entangled, highly similar feature clusters.

Ultimately, this study concludes that for datasets characterized by extreme intra-class environmental noise and severe inter-class visual similarities, deploying lightweight feature extractors even when paired with sophisticated machine learning heads is insufficient. To achieve the clinical-grade reliability and mathematical precision demanded by automated medication recognition systems, the deployment of deep, high-capacity architectures like ResNet50 is an absolute necessity.

CONCLUSION AND DISCUSSION

This study has comprehensively demonstrated the architectural requirements and structural limitations inherent in deploying deep learning for automated pill recognition. The findings explicitly validate that while lightweight architectures like MobileNetV2 are computationally efficient at extracting generalized feature vectors from high-

dimensional visual data, they fundamentally lack the depth required to resolve severe morphological ambiguities. A key insight highlighted during this research is the bottleneck effect observed in hybrid classification approaches. The attempt to augment the abstract features extracted by MobileNetV2 through statistical classifiers specifically Support Vector Machines (SVM) and k-Nearest Neighbors (kNN) proved that advanced boundary-drawing algorithms cannot compensate for low-resolution embeddings. Because the PCA-reduced feature space lacked fine-grained textural and geometric distinctions, the CNN+SVM (79.23%) and CNN+kNN (78.50%) frameworks failed to meaningfully surpass the baseline MobileNetV2 performance (79.22%), ultimately validating that a classifier's decision-making capacity is strictly bound by the quality of its input data.

The statistical results derived from the rigorous 5-fold cross-validation method provide compelling evidence regarding both the consistency of the developed systems and their definitive performance ceilings. While the narrow variance across different data subsets confirms that the hybrid and lightweight models avoided stochastic overfitting and remained structurally stable, their inability to break the 80% accuracy threshold reveals a critical shortfall in generalizability for highly similar medications (such as the persistent confusion between Bioflu and Neozept). Conversely, the deployment of the deep residual network, ResNet50, shattered this ceiling by achieving a benchmark accuracy of 90.18%. This robust performance academically and empirically validates that to meet the stringent requirements of the healthcare sector where the acceptable margin of error must be near zero high-capacity deep learning architectures are an absolute necessity over computationally constrained mobile networks.

Ultimately, this research underscores the vital role that the proper convergence of artificial intelligence and medical imaging technologies plays in enhancing medication safety. The "Drug Vision" project provides a clear architectural hierarchy tailored for pharmaceutical automation and individual medication management. Evaluating both the ethical framework and the technical realities of this study, the long-term objective is to transition this system into a real-time mobile application for patient assistance, supported by even more extensive training datasets. However, the rigorous analyses firmly conclude that relying on lightweight feature extractors or simple classifier swapping compromises diagnostic precision. By proving that the profound visual representation capabilities of deep residual architectures like ResNet50 are required to minimize medical errors, this study offers a reliable, high-precision technological foundation for the digital health ecosystem of tomorrow.

AUTHOR CONTRIBUTION STATEMENT

Each stage of the "Drug Vision" study was meticulously conducted in accordance with the principles of scientific merit and academic collaboration, with the entire process shaped by the active participation of the authors. During the conceptualization phase and hypothesis formulation, a collective brainstorming process was employed to clearly define the challenges of automated medication recognition in unconstrained environments and to determine optimal methodological solutions. In the data collection and preparation stages which represent the primary technical foundation of the research the standardization of the 10-class pharmaceutical dataset, image classification, and the design of essential preprocessing steps were executed under the joint coordination of the authors. Throughout this phase, detailed inspections were performed on the image data to maximize dataset quality, manage environmental noise, and ensure strict compliance with academic standards.

During the architectural design and technical coding phases, the implementation and comparative evaluation of deep residual networks (ResNet50), lightweight feature extractors (MobileNetV2, InceptionV3), and the hybrid classification frameworks (CNN+SVM and CNN+kNN) were executed utilizing a rigorous engineering approach. In the critical stages of algorithm optimization, hyperparameter tuning (including aggressive dropout regularization), and the execution of the rigorous 5-fold cross-validation tests, each author assumed direct responsibility for verifying the accuracy, stability, and reliability of the experimental outputs. The subsequent analysis of the statistical results, the interpretation of the ROC/AUC curves and confusion matrices, and the comparative evaluation of architectural bottlenecks were finalized collectively to guarantee the academic consistency of the project.

In the manuscript preparation phase, the structuring of the introduction, methodology, findings, and conclusion sections, as well as the generation of technical graphics (such as t-SNE visualizations and learning curves) and the integration of visual materials, were distributed equitably among the authors. The critical review of the final manuscript, rigorous editing, and the verification of scientific content were completed with the explicit approval of each contributor. All authors have reached a full consensus on the final version of the submitted manuscript and assume joint academic accountability for the integrity of the empirical data and the objectivity of the analyses presented herein. This comprehensive collaboration reflects not only the technical success of the project but also its execution with rigorous, interdisciplinary academic discipline.

STATEMENT OF SUPPORT AND THANKS

The successful completion of the "Drug Vision" project and its contribution to the academic literature were made possible through the technical, academic, and motivational support of numerous institutions and individuals. We would like to express our deepest gratitude to our esteemed instructor, whose rigorous guidance on coding standards, environment isolation, and project organization steered us through every stage of the research. Their strict mentorship and constructive warnings were the primary driving force in ensuring that this project evolved beyond a standard programming application to achieve genuine scientific depth and methodological discipline. Furthermore, we are deeply grateful to our institution for providing the foundational educational infrastructure utilized during the development and experimental phases.

We also owe a special debt of gratitude to the open-source communities and software developers who built the technical backbone of this study. The powerful tools and accessible environments provided by ecosystems such as Anaconda, Jupyter Lab, and Google Colab enabled our innovative concepts to materialize into tangible prototypes efficiently, optimizing our RAM usage and overcoming hardware constraints. Similarly, we acknowledge the invaluable contributions of the researchers who developed robust architectures like ResNet50 and MobileNetV2, making advanced deep learning accessible to the global scientific community. Without this collective technological foundation, evaluating complex hybrid pipelines (CNN+SVM, CNN+kNN) and achieving a 90.18% benchmark accuracy rate would not have been possible. Believing that knowledge multiplies when shared, we extend our sincere thanks to every individual contributing to this open ecosystem.

Finally, we would like to thank our families, friends, and our dormitory roommates for their unwavering support,

patience, and understanding throughout this intensive research process. During the long hours of debugging and the moments of mental fatigue inherent in academic research often extending late into the night in our shared living spaces their continuous moral support, observational dynamics, and shared camaraderie served as our greatest source of strength. This study is the product not only of technical achievement but also of disciplined collaboration and steadfast personal encouragement. We take this opportunity to express our sincere appreciation to everyone who contributed, directly or indirectly, to the realization of this project. We hope that this research serves as a valuable and beneficial contribution to the future of automated healthcare technologies.

CONFLICT OF INTEREST STATEMENT

All authors explicitly declare that there are no personal, financial, or institutional conflicts of interest that could compromise scientific objectivity or impartiality during the execution of the "Drug Vision" study, the analysis of the empirical results, or the drafting of this manuscript. The evaluation and presentation of all data obtained throughout the research process were conducted entirely within the framework of rigorous academic standards and engineering ethics. The authors hold no financial partnerships, consultancy agreements, stock ownership, or similar affiliations with the pharmaceutical companies producing the ten specific medications analyzed (including brands such as Biogesic, Neozep, and Bioflu), medical device manufacturers, or commercial entities operating in the field of artificial intelligence that could directly or indirectly influence the findings. This study was conceived as a completely independent academic endeavor and was completed without any external commercial or corporate funding.

Furthermore, it must be emphasized that the empirical results of this study are not intended to promote the commercial success of any specific product or brand, nor to disadvantage competitors; the primary objective is solely to advance the academic literature and support automated public health initiatives. The deep residual networks (ResNet50), lightweight feature extractors (MobileNetV2, InceptionV3), and the classical machine learning classifiers (SVM, kNN) employed in these comparative frameworks are entirely open-source, publicly accessible tools. Their selection and evaluation were based strictly on objective technical performance criteria. The authors confirm that they were not influenced by any external institution, corporation, or individual during the research or publication phases, and all conclusions presented in the article reflect their independent scientific views. This declaration of transparency is provided to maintain the trust of the readers and the scientific community, and to uphold the fundamental integrity of the research.

REFERENCES

This section presents a comprehensive review that establishes the technical and theoretical foundations of the "Drug Vision" study, reflecting the current state of the literature and the scientific rationale behind the comparative methodologies employed. All works cited throughout the manuscript were selected from pioneering research in deep learning, computer vision, medical imaging, and hybrid machine learning frameworks. In compiling the bibliography, emphasis was placed not only on projects specifically focused on automated pill classification systems but also on foundational academic publications that provide the theoretical infrastructure for deep residual networks (such as

ResNet50), lightweight feature extractors (MobileNetV2 and InceptionV3), transfer learning methodologies, and the rigorous regularization techniques central to this work. Furthermore, to ensure methodological transparency, the reference pool includes documentation for the core software libraries utilized such as the Python ecosystem, TensorFlow, Keras, and Scikit-learn alongside peer-reviewed studies demonstrating their practical applications in medical diagnostics.

The reference selection process encompassed a broad spectrum of literature, ranging from the historical evolution of image classification technologies to contemporary comparative analyses of standalone and hybrid deep learning pipelines. This approach synthesizes various perspectives, spanning classic papers on the boundary-drawing efficacy of Support Vector Machines (SVM) and the proximity-based mechanics of k-Nearest Neighbors (kNN) in high-dimensional feature spaces, to recent articles addressing the structural bottlenecks of lightweight neural networks when resolving complex morphological overlaps. The review encompasses CNN-based research conducted on comparable multi-class medication datasets, as well as foundational studies addressing broader topics such as hyperparameter optimization, cross-validation stability, and the deployment of AI in healthcare ecosystems. This extensive bibliographic framework demonstrates that the present research is not merely an isolated engineering application, but rather a study deeply rooted in the multi-layered accumulation of knowledge within the artificial intelligence literature, aiming to bridge the gap between theoretical computer vision and real-world pharmaceutical automation.

The fundamental academic sources utilized during the design, coding, and reporting phases of this study are listed below:

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AUTHOR GUIDELINES

This article, alongside the software infrastructure developed for the "Drug Vision" study, has been structured in strict accordance with rigorous engineering disciplines and academic standards. The guiding principles adopted throughout the writing and development phases prioritize the transparency of technical processes and the consistency of the empirical results. The experimental pipeline was constructed utilizing TensorFlow and Keras, powerful libraries within the Python ecosystem. Every step, from the physical standardization of the dataset to the precise tuning of hyperparameters such as aggressive dropout regularization has been meticulously documented. During the implementation phase, the memory-efficient processing of complex image data via the ImageDataGenerator and its subsequent conversion into a format suitable for the models' input layers established the foundation of the technical methodology. As explicitly noted in the project's documentation, the distribution and sample size of each medication class (Alaxan, Bactidol, Bioflu, Biogesic, DayZinc, Decolgen, Fish Oil, Kremil S, Medicol, and Neozepe) were carefully monitored to ensure unbiased training across all evaluated architectures.

The most critical stage of the development process the design and evaluation of the model architectures was built upon a comparative framework testing deep residual networks (ResNet50) against lightweight feature extractors (MobileNetV2, InceptionV3). To leverage robust pre-trained weights from the ImageNet dataset, the foundational convolutional layers were utilized via transfer learning. Furthermore, a hybrid strategy was implemented to assess whether traditional mathematical classifiers could artificially enhance lightweight feature embeddings. The abstract, high-dimensional features extracted by MobileNetV2 were subjected to Principal Component Analysis (PCA) to

reduce dimensionality, subsequently serving as input signatures for Support Vector Machine (SVM) and k-Nearest Neighbors (kNN) algorithms. Parameter optimizations, such as selecting the Radial Basis Function (RBF) kernel for the SVM and tuning the distance metrics for kNN, were refined based on rigorous experimental outcomes. This guide thoroughly explains the logical sequence underlying these architectural choices, illustrating both the superior discriminative capabilities of deep residual networks and the structural bottlenecks encountered during the hybridization process.

Finally, the validation and reporting phase of the project was solidified using a rigorous 5-fold cross-validation method to satisfy the highest requirements of academic reliability. The computational loop structures embedded within the codebase serve as a practical technical guide, demonstrating how each data segment was systematically partitioned during both the training and testing phases to eliminate stochastic deviations and prevent overfitting. The resulting confusion matrices, class-specific ROC/AUC curves, and comparative performance graphs have been incorporated into the main text as fundamental empirical evidence, validating both the accuracy of ResNet50 and the limitations of the lightweight models. Ultimately, this guideline affirms that the disciplined roadmap followed throughout the study is more than a simple sequence of programming scripts; every step is deeply rooted in academic standards, collectively serving as a comprehensive, comparative application manual for future research in automated medical image analysis.